
Integration Manual

for MPC560XB LIN Driver

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Chapter 1

Revision History

Table 1-1. Revision History

Revision	Date	Author	Description
1.0	31-Jan-2012	Giuseppe Stefano Fazio	First version for Beta 0.9.0 release
1.1	06-May-2013	Giuseppe Stefano Fazio	Updated the manual for Bolero RTM 1.0.0 release
1.2	06-May-2014	Giuseppe Stefano Fazio	Updated the manual for Bolero RTM 1.0.1 release



Chapter 2

Introduction

This integration manual describes the integration requirements for Lin Driver for MPC560XB microcontrollers.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of Freescale Semiconductor .

Table 2-1. MPC560XB Derivatives

Freescale Semiconductor	mpc5601dxlh_lqfp64 mpc5601dxll_lqfp100 mpc5602dxlh_lqfp64 mpc5602dxll_lqfp100 mpc5602bxlh_lqfp64 mpc5602bxll_lqfp100 mpc5602bxlq_lqfp144 mpc5602cxlh_lqfp64 mpc5602cxll_lqfp100 mpc5603bxlh_lqfp64 mpc5603bxll_lqfp100 mpc5603bxlq_lqfp144 mpc5603cxlh_lqfp64 mpc5603cxll_lqfp100 mpc5604bxlh_lqfp64 mpc5604bxll_lqfp100 mpc5604bxlq_lqfp144 mpc5604bxmg_mapbga208 mpc5604cxlh_lqfp64 mpc5604cxll_lqfp100 mpc5605bxll_lqfp100 mpc5605bxlq_lqfp144 mpc5605bxlu_lqfp176 mpc5606bkll_lqfp100 mpc5606bxlq_lqfp144 mpc5606bxlu_lqfp176 mpc5607bxlu_lqfp176 mpc5607bxmg_mapbga208
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All of the above microcontroller devices are collectively named as MPC560XB .

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About this Manual

This Technical Reference employs the following typographical conventions:

Boldface type: Bold is used for important terms, notes and warnings.

Italic font: Italic typeface is used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

2.4 Acronyms and Definitions

Table 2-2. Acronyms and Definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
AUTOSAR	Automotive Open System Architecture
BSMI	Basic Software Make file Interface
C/CPP	C and C++ Source Code
DEM	Diagnostic Event Manager
DET	Development Error Tracer
EcuM	ECU state Manager
GUI	Graphical User Interface
ISR	Interrupt Service Routine
LIN	Local Interconnect Network
MCU	Micro Controller Unit
N/A	Not Applicable
OS	Operating System
PB Variant	Post Build Variant
PC Variant	Pre Compile Variant
VLE	Variable Length Encoding

2.5 Reference List

Table 2-3. Reference List

#	Title	Version
1	AUTOSAR 4.0 Rev0003Lin Driver Software Specification Document.	V1.5.0
2	MPC5607B Reference Manual	Rev. 7.2, May 2012
3	Bolero 1.5M cut 2.2 Errata Sheet (MPC5607B_1M03Y)	Rev. 12Dec2013
4	MPC5604B Reference Manual	Rev. 8.2, September 2013
5	Bolero 512k cut 2.3 Errata Sheet (MPC5604B_2M27V)	Rev. 26 FEB 2014
6	5606B Reference Manual	Rev. 2, May 2014
7	5602D Reference Manual	Rev. 4.2, Dec 2013
8	Bolero 256k cut 1.1 Errata Sheet (MPC5602D_1M18Y)	Rev. , 12 DEC 2013

Chapter 3

Building the Driver

This section describes the source files and various compilers, linker options used for building the Autosar Lin driver for Freescale Semiconductor MPC560XB . It also explains the EB Tresos Studio plugin setup procedure.

3.1 Build Options

The Lin driver files are compiled using

- Green Hills MULTI v6.1.4
- Windriver DIAB Rel DIAB_5_9_2_0-FCS_20120725_234511
- Freescale CodeWarrior for MPC5xxx v2.10 - compiler Version 4.3 build 224
- HighTec v4.6.4.0, Compiler gcc v4.6.3

The compiler, linker flags used for building the driver are explained below:

Note

The TS_T2D13M10I1R0 plugin name is composed as follow:

TS_T = Target_Id

D = Derivative_Id

M = SW_Version_Major

I = SW_Version_Minor

R = Revision

(i.e. Target_Id = 2 identifies PowerPC architecture and
Derivative_Id = 13 identifies the MPC560XB)

3.1.1 CW Compiler/Linker/Assembler Options

Table 3-1. Compiler Options

Option	Description
-proc Zen	Generates and links object code for Zen processor. The compiler uses unsigned as the default parameter for the -char switch
-lang c	Expects source code to conform to the language specified by the ISO/IEC 9899-1990 ("C90") standard
-opt all	This option is selected all optimization (the same as -opt speed,level=4,intrinsics,noframe)
-common off	Disables moving uninitialized data into a common section
-sdatathreshold 0	Specifies the threshold size (in bytes) for an item considered by the linker to be small data. (The linker stores small data items in the Small Data address space. The compiler can generate faster code to access this data.)
-sdata2threshold 0	Specifies the threshold size (in bytes) for an item considered by the linker to be small constant data. (The linker stores small constant data items in the Small Constant Data address space.)
-vle	Tells the compiler and linker to generate and lay out Variable Length Encoded (VLE) instructions, available on Zen variants of Power Architecture processors
-use_lmw_stmw on	Enables the use of multiple load and store instructions for function prologues and epilogues
-ppc_asm_to_vle	Converts regular Power Architecture assembler mnemonics to equivalent VLE (Variable Length Encoded) assembler mnemonics in the inline assembler
-Cpp_exceptions off	When on, generates executable code for C++ exceptions. When off, generates smaller, faster executable code
-func_align 4	Specifies alignment of functions in executable code
-sym dwarf-2,full	Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format)
-gdwarf-2	Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format). The linker ignores debugging information that is not in the Dwarf 1, Dwarf 2 format
-w on	Turns on most warning messages
-r	Compiler should expect function prototypes
-w undefmacro	Issues warning messages on the use of undefined macros in #if and #elif conditionals
-char unsigned	Controls the default sign of the char data type: char data items are unsigned
-nosyspath	Performs a search of both the user and system paths, treating #include statements of the form #include xyz the same as the form #include "xyz"
-fp none	No floating point code generation
-DAUTOSAR_OS_NOT_USED	-D defines a preprocessor symbol and optionally can set it to a value. AUTOSAR_OS_NOT_USED: By default in the package, the drivers are compiled to be used without Autosar OS. If the drivers are used with Autosar OS, the compiler option '-DAUTOSAR_OS_NOT_USED' must be removed from project options
-DMWERKS	-D defines a preprocessor symbol and optionally can set it to a value. This one defines the CW preprocessor symbol.

Table 3-2. Assembler Options

Option	Description
-proc Zen	Generates and links object code for Zen processor. The compiler uses unsigned as the default parameter for the -char switch

Table continues on the next page...

Table 3-2. Assembler Options (continued)

Option	Description
-vle	Tells the compiler and linker to generate and lay out Variable Length Encoded (VLE) instructions, available on Zen variants of Power Architecture processors
-sym dwarf-2,full	Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format)
-gdwarf-2	Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format). The linker ignores debugging information that is not in the Dwarf 1, Dwarf 2 format.
-DMWERKS	-D defines a preprocessor symbol and optionally can set it to a value. This one defines the CW preprocessor symbol.

Table 3-3. Linker Options

Option	Description
-proc Zen	Generates and links object code for Zen processor. The compiler uses unsigned as the default parameter for the -char switch
-code_merging all	Removes duplicated functions to reduce object code size
-far_near_addressing	Simplifies address computations to reduce object code size and improve performance
-vle_enhance_merging	Removes duplicated functions that are called by functions that use VLE instructions to reduce object code size
-listdwarf	DWARF debugging information in the linker's map file
-sym dwarf-2,full	Generate DWARF-2-conforming debugging information (Debug With Arbitrary Record Format)
-char unsigned	Controls the default sign of the char data type: char data items are unsigned.

3.1.2 DIAB Compiler/Linker/Assembler Options

Table 3-4. Compiler Options

Option	Description
-Xdialect-ansi	Follow the ANSI C standard with some additions
-XO	Enables extra optimizations to produce highly optimized code
-g	Generate symbolic debugger information and disable some optimizations.
-Xsize-opt	Optimize for size rather than speed when there is a choice
-Xsmall-data=0	Set Size Limit for "small data" Variables to zero.
-Xaddr-sconst=0x11	Specify addressing for constant static and global variables with size less than or equal to -Xsmall-const to far-absolute.
-Xaddr-sdata=0x11	Specify addressing for non-constant static and global variables with size less than or equal to -Xsmall-data in size to far-absolute.
-Xno-common	Disable use of the "COMMON" feature so that the compiler or assembler will allocate each uninitialized public variable in the .bss section for the module defining it, and the linker will require exactly one definition of each public variable
-Xnested-interrupts	Allow nested interrupts

Table continues on the next page...

Table 3-4. Compiler Options (continued)

Option	Description
-Xdebug-dwarf2	Generate symbolic debug information in dwarf2 format
-Xdebug-local-all	Force generation of type information for all local variables
-Xdebug-local-cie	Create common information entry per module
-Xdebug-struct-all	Force generation of type information for all typedefs, struct, union and class types
-Xforce-declarations	Generates warnings if a function is used without a previous declaration
-ee1481	Generate an error when the function was used before it has been declared
-Xmacro-undefined-warn	Generates a warning when an undefined macro name occurs in a #if preprocessor directive
-W:as,-l	Pass the option "-l" (lower case letter L) to the assembler to get an assembler listing file
-Wa,-Xisa-vle	Instruct the assembler to expect and assemble VLE (Variable Length Encoding) instructions rather than BookE instructions.
-DAUTOSAR_OS_NOT_USED	-D defines a preprocessor symbol and optionally can set it to a value. AUTOSAR_OS_NOT_USED: By default in the package, the drivers are compiled to be used without Autosar OS. If the drivers are used with Autosar OS, the compiler option '-DAUTOSAR_OS_NOT_USED' must be removed from project options
-DDIAB	-D defines a preprocessor symbol and optionally can set it to a value. This one defines the DIAB preprocessor symbol.
-Xforce-prototypes	-Generate warnings if a function is used without a previous prototype declaration.
-tPPCE200Z0VEN:simple	Sets target processor to PPCE200Z0, generates ELF using EABI conventions, No floating point support (minimizes the required runtime), selects simple environment settings for Startup Module and Libraries.

Table 3-5. Assembler Options

Option	Description
-tPPCE200Z0VEN:simple	Sets target processor to PPCE200Z0, generates ELF using EABI conventions, No floating point support (minimizes the required runtime), selects simple environment settings for Startup Module and Libraries
-g	Dump the symbols in the global symbol table in each archive file.
-Xisa-vle	Expect and assemble VLE (Variable Length Encoding) instructions rather than Book E instructions. The default code section is named .text_vle instead of .text, and the default code section fill "character" is set to 0x44444444 instead of 0. The .text_vle code section will have ELF section header flags marking it as VLE code, not Book E code.

Table 3-6. Linker Options

Option	Description
-tPPCE200Z0VEN:simple	Sets target processor to PPCE200Z0, generates ELF using EABI conventions, No floating point support (minimizes the required runtime), selects simple environment settings for Startup Module and Libraries
-Xelf	Generates ELF object format for output file
-m6	Generates a detailed link map and cross reference table
-lc	Specifies to linker to search for libc.a
-Xlibc-old	Enables usage of legacy (pre-release 5.6) libraries

3.1.3 GHS Compiler/Linker/Assembler Options

Table 3-7. Compiler Options

Option	Description
-cpu=ppc560xb	Selects target processor: ppc560xb
-ansi	Specifies ANSI C with extensions. This mode extends the ANSI X3.159-1989 standard with certain useful and compatible constructs.
-noSPE	Disables the use of SPE and vector floating point instructions by the compiler.
-Ospace	Optimize for size
-sda=0	Enables the Small Data Area optimization with a threshold of 0.
-vle	Enables VLE code generation
-dual_debug	Enables the generation of DWARF, COFF, or BSD debugging information in the object file
-G	Generates source level debugging information and allows procedure call from debugger's command line.
--no_exceptions	Disables support for exception handling
-Wundef	Generates warnings for undefined symbols in preprocessor expressions
-Wimplicit-int	Issues a warning if the return type of a function is not declared before it is called
-Wshadow	Issues a warning if the declaration of a local variable shadows the declaration of a variable of the same name declared at the global scope, or at an outer scope
-Wtrigraphs	Issues a warning for any use of trigraphs
--prototype_errors	Generates errors when functions referenced or called have no prototype
--incorrect_pragma_warnings	Valid #pragma directives with wrong syntax are treated as warnings
-noslashcomment	C++ like comments will generate a compilation error
-preprocess_assembly_files	Preprocesses assembly files
--short_enum	Store enumerations in the smallest possible type
--no_commons	Allocates uninitialized global variables to a section and initializes them to zero at program startup.
-c	Produces an object file (called input-file.o) for each source file.
-DAUTOSAR_OS_NOT_USED	-D defines a preprocessor symbol and optionally can set it to a value. AUTOSAR_OS_NOT_USED: By default in the package, the drivers are compiled to be used without Autosar OS. If the drivers are used with Autosar OS, the compiler option '-DAUTOSAR_OS_NOT_USED' must be removed from project options
-DUSE_SW_VECTOR_MODE	-D defines a preprocessor symbol and optionally can set it to a value. USE_SW_VECTOR_MODE: By default in the package, drivers are compiled to be used with interrupt controller configured to be in hardware vector mode. In case of AUTOSAR_OS_NOT_USED, the compiler option "-DUSE_SW_VECTOR_MODE" must be added to the list of compiler options to be used with interrupt controller configured to be in software vector mode.
-DDISABLE_MCAL_INTERMODULE_ASR_CHECK	-D defines a preprocessor symbol to disable the inter-module version check for AR_RELEASE versions. DISABLE_MCAL_INTERMODULE_ASR_CHECK: By default in the package, drivers are compiled to perform the inter-module version check as per Autosar BSW004. When the inter-module version check needs to be disabled then the DISABLE_MCAL_INTERMODULE_ASR_CHECK global define must be added to the list of compiler options.
-DGHS	-D defines a preprocessor symbol and optionally can set it to a value. This one defines the GHS preprocessor symbol.

Table 3-8. Assembler Options

Option	Description
-cpu=ppc560xb	Selects target processor: ppc560xb

Table 3-9. Linker Options

Option	Description
-cpu=ppc560xb	Selects target processor: ppc560xb
-nostartfiles	Do not use Start files.
-vle	Enables VLE code generation
-linker_warnings	Display linker warnings

3.1.4 HT Compiler/Linker/Assembler Options

Table 3-10. Compiler Options

Option	Description
-Os	Optimize for size. '-Os' enables all '-O2' optimizations that do not typically increase code size. It also performs further optimizations designed to reduce code size.
-msdata=eabi	On System V.4 and embedded PowerPC systems, put small initialized const global and static data in the '.sdata2' section, which is pointed to by register r2. Put small initialized non-const global and static data in the '.sdata' section, which is pointed to by register r13. Put small uninitialized global and static data in the '.sbss' section, which is adjacent to the '.sdata' section.
-G1000	On embedded PowerPC systems, put global and static items less than or equal to num bytes into the small data or bss sections instead of the normal data or bss section.
-fshort-double	Use the same size for double as for float.
-mno-spe	This option used to disable spe extension (Like -noSPE on ghs compiler).
-mregnames	On System V.4 and embedded PowerPC systems do (do not) emit register names in the assembly language output using symbolic forms
-g3	Request debugging information of Level 3. Includes extra information, such as all the macro definitions present in the program.
-Wall	This enables all the warnings about constructions that some users consider questionable, and that are easy to avoid (or modify to prevent the warning), even in conjunction with macros.
-mvle	Enable support of VLE
--mcpu=z425n3	Support of e200 Z425n3 core
-DAUTOSAR_OS_NOT_USED	-D defines a preprocessor symbol and optionally can set it to a value. AUTOSAR_OS_NOT_USED: By default in the package, the drivers are compiled to be used without Autosar OS. If the drivers are used with Autosar OS, the compiler option '-DAUTOSAR_OS_NOT_USED' must be removed from project options
-DUSE_SW_VECTOR_MODE	-D defines a preprocessor symbol and optionally can set it to a value. USE_SW_VECTOR_MODE: By default in the package, drivers are compiled to be used with interrupt controller configured to be in hardware vector mode. In case of AUTOSAR_OS_NOT_USED, the compiler option "-DUSE_SW_VECTOR_MODE" must be

Table 3-10. Compiler Options

Option	Description
	added to the list of compiler options to be used with interrupt controller configured to be in software vector mode.

Table 3-11. Assembler Options

Option	Description
-c	Compile or assemble the source files, but do not link. The linking stage simply is not done. The ultimate output is in the form of an object file for each source file.
-Wa,--gdwarf2	Produce debugging information in DWARF format.
-msdata=eabi	Put small initialized const global and static data in the .sdata2 section, which is pointed to by register r2. Put small initialized non-const global and static data in the .sdata section, which is pointed to by register r13. Put small uninitialized global and static data in the .sbss section, which is adjacent to the .sdata section.
-mregnames	Do emit register names in the assembly language output using symbolic forms.
-mno-spe	This option used to disable spe extension (Like --noSPE on ghs compiler).
-mcpu=z425n3	Support of e200 z425n3 core

Table 3-12. Linker Options

Option	Description
-N	Set the text and data sections to be readable and writable. Also, do not page-align the data segment. If the output format supports Unix style magic numbers, mark the output as OMAGIC.
--no-undefined	Normally when a symbol has an undefined version, the linker will ignore it. This option disallows symbols with undefined version and a fatal error will be issued instead.
--warn-common	Normally ppc-vle-ld will give a warning if it finds an incompatible library during a library search. This option silences the warning.
--error-unresolved-symbols	This restores the linker's default behaviour of generating errors when it is reporting unresolved symbols.
-nocrt0	library crt0 from compiler library not used, it uses custom crt0 file
-nostartfiles	no start file from compiler library is used
-Wl	If the linker is being invoked indirectly, via a compiler driver (e.g. 'gcc') then all the linker command line options should be prefixed by '-Wl,'Used in combination with -no-warn-flags, --no-undefined, --warn-common and --error-unresolved-symbols
--no-warn-flags	Used in combination with -Wl disable linker warnings. See previous
-T	The -T option specifies the name of your own script ie sprig.link. The -c is a synonym of the -T option

3.2 Files required for Compilation

This section describes the include files required to compile, assemble (if assembler code) and link the Lin driver for MPC560XB microcontrollers.

To avoid integration of incompatible files, all the include files from other modules shall have the same AR_MAJOR_VERSION and AR_MINOR_VERSION, i.e. only files with the same AUTOSAR major and minor versions can be compiled.

Lin Files

- ..\Lin_TS_T2D13M10I1R0\src\Lin.c
- ..\Lin_TS_T2D13M10I1R0\src\Lin_Irq.c
- ..\Lin_TS_T2D13M10I1R0\src\Lin_LLD.c
- ..\Lin_TS_T2D13M10I1R0\src\Lin_NonASR.c
- ..\Lin_TS_T2D13M10I1R0\src\Lin_Linflex_LLD.c

- ..\Lin_TS_T2D13M10I1R0\include\Lin.h
- ..\Lin_TS_T2D13M10I1R0\include\Lin_NonASR.h
- ..\Lin_TS_T2D13M10I1R0\include\Lin_LLD.h
- ..\Lin_TS_T2D13M10I1R0\include\Lin_Linflex_LLD.h
- ..\Lin_TS_T2D13M10I1R0\include\Reg_eSys_LINFlex.h
- ..\Lin_TS_T2D13M10I1R0\include\Lin_Linflex_LLD_CfgEx.h

Lin Generated Files

- Lin_PBcfg.c (For PB Variant) - This file should be generated by the user using a configuration tool for compilation (it contains initialization values for Pre Compile variant).
- Lin_Cfg.c (For PC Variant) - This file should be generated by the user using a configuration tool for compilation (it contains initialization values for Post Build variant).
- Lin_Cfg.h (For PC and PB Variants) - This file should be generated by the user using a configuration tool for compilation (common file containing data and structure definition).

Files from Base common folder

- ..\Base_TS_T2D13M10I1R0\generate_PC\include\modules.h
- ..\Base_TS_T2D13M10I1R0\generate_PC\include\Soc_Ips.h
- ..\Base_TS_T2D13M10I1R0\include\Cer.h
- ..\Base_TS_T2D13M10I1R0\include\Compiler.h
- ..\Base_TS_T2D13M10I1R0\include\Compiler_Cfg.h
- ..\Base_TS_T2D13M10I1R0\include\ComStack_Cfg.h
- ..\Base_TS_T2D13M10I1R0\include\ComStack_Types.h
- ..\Base_TS_T2D13M10I1R0\include\Lin_GeneralTypes.h
- ..\Base_TS_T2D13M10I1R0\include\Mcal.h

- ..\Base_TS_T2D13M10I1R0\include\MemMap.h
- ..\Base_TS_T2D13M10I1R0\include\Platform_Types.h
- ..\Base_TS_T2D13M10I1R0\include\Reg_eSys.h
- ..\Base_TS_T2D13M10I1R0\include\Reg_Macros.h
- ..\Base_TS_T2D13M10I1R0\include\Std_Types.h

Files from Dem folder:

- ..\Dem_TS_T2D13M10I1R0\generate_PC\include\Dem_IntErrId.h
- ..\Dem_TS_T2D13M10I1R0\include\Dem.h
- ..\Dem_TS_T2D13M10I1R0\include\Dem_Types.h

Files from Det folder:

- ..\Det_TS_T2D13M10I1R0\include\Det.h

Files from EcuM folder:

- ..\EcuM_TS_T2D13M10I1R0\generate_PC\include\EcuM_Cfg.h
- ..\EcuM_TS_T2D13M10I1R0\include\EcuM.h
- ..\EcuM_TS_T2D13M10I1R0\include\EcuM_Cbk.h

Files from Rte folder:

- Rte_TS_T2D13M10I1R0\include\SchM_Adc.h
- Rte_TS_T2D13M10I1R0\src\SchM_Adc.c
- Rte_TS_T2D13M10I1R0\include\SchM_Can.h
- Rte_TS_T2D13M10I1R0\src\SchM_Can.c
- Rte_TS_T2D13M10I1R0\include\SchM_Dio.h
- Rte_TS_T2D13M10I1R0\src\SchM_Dio.c
- Rte_TS_T2D13M10I1R0\include\SchM_Eth.h
- Rte_TS_T2D13M10I1R0\src\SchM_Eth.c
- Rte_TS_T2D13M10I1R0\include\SchM_Fls.h
- Rte_TS_T2D13M10I1R0\src\SchM_Fls.c
- Rte_TS_T2D13M10I1R0\include\SchM_Gpt.h
- Rte_TS_T2D13M10I1R0\src\SchM_Gpt.c
- Rte_TS_T2D13M10I1R0\include\SchM_Icu.h
- Rte_TS_T2D13M10I1R0\src\SchM_Icu.c
- Rte_TS_T2D13M10I1R0\include\SchM_Mcu.h
- Rte_TS_T2D13M10I1R0\src\SchM_Mcu.c
- Rte_TS_T2D13M10I1R0\include\SchM_Port.h
- Rte_TS_T2D13M10I1R0\src\SchM_Port.c
- Rte_TS_T2D13M10I1R0\include\SchM_Pwm.h
- Rte_TS_T2D13M10I1R0\src\SchM_Pwm.c
- Rte_TS_T2D13M10I1R0\include\SchM_RamTst.h
- Rte_TS_T2D13M10I1R0\src\SchM_RamTst.c
- Rte_TS_T2D13M10I1R0\include\SchM_Spi.h

- Rte_TS_T2D13M10I1R0\src\SchM_Spi.c
- Rte_TS_T2D13M10I1R0\include\SchM_Wdg.h
- Rte_TS_T2D13M10I1R0\src\SchM_Wdg.c

3.3 Setting up the Plug-ins

The Lin driver was designed to be configured by using the EB Tresos Studio (version Elektrobit Tresos Release 14.2.1 (b140128-1223) or later.)

Location of various files inside the LIN module folder:

- VSMD (Vendor Specific Module Definition) file in EB tresos Studio XDM format:
 - ..\Base_TS_T2D13M10I1R0\config\Base.xdm
 - ..\EcuM_TS_T2D13M10I1R0\config\EcuM.xdm
 - ..\Lin_TS_T2D13M10I1R0\config\Lin.xdm
 - ..\Mcu_TS_T2D13M10I1R0\config\Mcu.xdm
 - ..\Resource_TS_T2D13M10I1R0\config\Resource.xdm
- VSMD (Vendor Specific Module Definition) file(s) in AUTOSAR compliant EPD format:
 - ..\Base_TS_T2D13M10I1R0\autosar\Base.epd
 - ..\EcuM_TS_T2D13M10I1R0\autosar\EcuM.epd
 - ..\Lin_TS_T2D13M10I1R0\autosar\Lin.epd
 - ..\Mcu_TS_T2D13M10I1R0\autosar\Mcu.epd
 - ..\Resource_TS_T2D13M10I1R0\autosar\Resource.epd
- Code Generation Templates for Pre-Compile time configuration parameters:
 - ..\Lin_TS_T2D13M10I1R0generate_PC\src\Lin_Cfg.c
 - ..\Lin_TS_T2D13M10I1R0generate_PC\include\Lin_Cfg.h
 - ..\Lin_TS_T2D13M10I1R0generate_PC\Lin_VersionCheck_Inc.m
 - ..\Lin_TS_T2D13M10I1R0generate_PC\Lin_VersionCheck_Src.m
 - ..\Lin_TS_T2D13M10I1R0generate_PC\Lin_BaudRate_Comp.m
- Code Generation Templates for Post-Build time configuration parameters:
 - ..\Lin_TS_T2D13M10I1R0generate_PB\src\Lin_PBcfg.c
 - ..\Lin_TS_T2D13M10I1R0generate_PB\Lin_VersionCheck_Src_PB.m
 - ..\Lin_TS_T2D13M10I1R0generate_PB\Lin_BaudRate_Comp.m

Steps to generate the configuration:

1. Copy the module folders Lin_TS_T2D13M10I1R0, Base_TS_T2D13M10I1R0, Resource_TS_T2D13M10I1R0, EcuM_TS_T2D13M10I1R0, Mcu_TS_T2D13M10I1R0 into the Tresos plugins folder.

2. Set the desired Tresos Output location folder for the generated sources and header files.
3. Use the EB tresos Studio GUI to modify ECU configuration parameters values.
4. Generate the configuration files.

Dependencies

- **MCU** is required to use System Clock when clock source is used as Peripheral clock source to generate LIN Segment values.
- **RESOURCE** is required to select processor derivative. Current Lin driver has support for the following derivatives, everyone having attached a Resource file:
 mpc5601dxlh_lqfp64 mpc5601dxll_lqfp100 mpc5602dxlh_lqfp64
 mpc5602dxll_lqfp100 mpc5602bxlh_lqfp64 mpc5602bxll_lqfp100
 mpc5602bxlq_lqfp144 mpc5602cxlh_lqfp64 mpc5602cxll_lqfp100
 mpc5603bxlh_lqfp64 mpc5603bxll_lqfp100 mpc5603bxlq_lqfp144
 mpc5603cxlh_lqfp64 mpc5603cxll_lqfp100 mpc5604bxlh_lqfp64
 mpc5604bxll_lqfp100 mpc5604bxlq_lqfp144 mpc5604bxmg_mapbga208
 mpc5604cxlh_lqfp64 mpc5604cxll_lqfp100 mpc5605bxll_lqfp100
 mpc5605bxlq_lqfp144 mpc5605bxlu_lqfp176 mpc5606bkll_lqfp100
 mpc5606bxlq_lqfp144 mpc5606bxlu_lqfp176 mpc5607bxlu_lqfp176
 mpc5607bxmg_mapbga208 .
- **ECUM** is required for selecting the reference to the wakeup source for every Lin controller.
- **DET** is required for signaling the development error detection (parameters out of range, null pointers, etc).
- **DEM** is required for signaling the production error detection (hardware failure, etc).

Resource Parameters Configuration

1. **Lin.LinGlobalConfig.LinChannel** - number of maximum available LINFlex controllers on chip.
2. **Lin.LinGlobalConfig.LinChannel.LinHwChannel** - list of available LinFlex controllers on chip (LinHWCh_0, LinHWCh_1, etc).
3. **LinExternalWKPUSupport** - TRUE if wake up support needs to be configured.
4. **LinExternalWKPUSupportChannelID** - External Wake up channel(s) assigned for each LIN HW channel.

Chapter 4

Function calls to module

4.1 Function Calls during Start-up

Lin shall be initialized during STARTUP phase of EcuM initialization. The API to be called for this is `Lin_Init()`. The MCU module should be initialized before the Lin is initialized. The Lin driver does not need OS Support except for ISR's. Hence, can be initialized either in STARTUP1 or STARTUP2 phase of EcuM initialization. This depends on the implementation, desired duration for STARTUP1 & Target hardware design. The LIN module shall be initialized by `Lin_Init(<&Lin_Configuration>)` service call during the start-up before the LIN peripherals are used. Please note that GPIO pins used for connection of LIN physical layer have to be properly assigned to desired LINFlex module prior the LIN initialization: so the MCU and PORT modules shall be initialized before LIN is initialized. After the LIN module is initialized each LIN channel have to be initialized as well before using it. This is also done by the `Lin_Init(<&Lin_Configuration>)` service.

4.2 Function Calls during Shutdown

There is no shutdown specific procedure for Lin driver. LIN driver can go to sleep mode using the following API services:

`Lin_GoToSleepInternal(LIN_CHANNEL)`: which put the LIN driver into sleep mode without sending of Go-to-sleep command over the bus.

`Lin_GoToSleep(LIN_CHANNEL)`: which put the LIN driver into sleep mode and send a Go-to-sleep command over the bus.

4.3 Function Calls during Wake-up

LIN driver supports the transmission of wake up command via the LIN bus.

For this purposes the `Lin_Wakeup(LIN_CHANNEL)` API service may be used.

External Wakeup:

The Lin driver supports external wake-up from the bus in 2 modes:

If the channel is configured with "wake-up support", upon wakeup detection on Rx pin of the configured LINFlex channel the ISR "`LINFlex_Tx_Rx_IRQHandler(LIN channel)`" will be executed (based on the LIN Channel configured) to wake-up Lin Driver.

If the channel is not configured with "wake-up support", the Lin stack may call `Lin_CheckWakeup(LIN_CHANNEL)` service API in order to identify if a slave on the Lin bus issued a wake-up request. In case such a request is identified then, it is the Lin stack responsibility to wake up the channel and process the slave request.

Chapter 5

Module requirements

5.1 Exclusive areas to be defined in BSW scheduler

None.

5.2 Peripheral Hardware Requirements

The LIN physical interface should be connected to the LIN module pins in order to get the LIN bus voltage levels.

5.3 ISR to configure within OS – dependencies

The interrupt service routines are implemented using ISR macro.

The ISR macro implementation depends on the MCAL environment used:

1. Without OS and INTC used in software vector mode:

```
#define ISR(IsrName) void IsrName(void)
```

2. Without OS and INTC used in hardware vector mode:

```
#define ISR(IsrName) INTERRUPT_FUNC void IsrName(void)
```

3. With Freescale OS:

```
#define ISR(IsrName) void OS_isr_##IsrName()
```

The following ISR's are used by the LIN driver:

Table 5-1. LIN ISR's For LINFlex channel

ISR Name	Hardware interrupt vector
Lin_LLD_IsrRx_LINFlex_0	79
Lin_LLD_IsrTx_LINFlex_0	80
Lin_LLD_IsrError_LINFlex_0	81
Lin_LLD_IsrRx_LINFlex_1	99
Lin_LLD_IsrTx_LINFlex_1	100
Lin_LLD_IsrError_LINFlex_1	101
Lin_LLD_IsrRx_LINFlex_2	119
Lin_LLD_IsrTx_LINFlex_2	120
Lin_LLD_IsrError_LINFlex_2	121
Lin_LLD_IsrRx_LINFlex_3	122
Lin_LLD_IsrTx_LINFlex_3	123
Lin_LLD_IsrError_LINFlex_3	124
Lin_LLD_IsrRx_LINFlex_4	187
Lin_LLD_IsrTx_LINFlex_4	188
Lin_LLD_IsrError_LINFlex_4	189
Lin_LLD_IsrRx_LINFlex_5	199
Lin_LLD_IsrTx_LINFlex_5	200
Lin_LLD_IsrError_LINFlex_5	201
Lin_LLD_IsrRx_LINFlex_6	216
Lin_LLD_IsrTx_LINFlex_6	217
Lin_LLD_IsrError_LINFlex_6	218
Lin_LLD_IsrRx_LINFlex_7	224
Lin_LLD_IsrTx_LINFlex_7	225
Lin_LLD_IsrError_LINFlex_7	226
Lin_LLD_IsrRx_LINFlex_8	227
Lin_LLD_IsrTx_LINFlex_8	228
Lin_LLD_IsrError_LINFlex_8	229
Lin_LLD_IsrRx_LINFlex_9	230
Lin_LLD_IsrTx_LINFlex_9	231
Lin_LLD_IsrError_LINFlex_9	232

For each interrupts name is added a prefix (ISR) that depends by OS:

- For MCAL without OS, ISR() macro is defined as “OSISR_”.
- For Freescale OS, the ISR() macro is defined as “OS_isr_”.
- For EB OS, the ISR() macro is defined as “OS_ISR_”.

NOTE:

1. The type number and interrupt vectors of hardware channels are specific for each platform. See the documentation for details.

2. In case of AUTOSAR_OS_NOT_USED and the user wants to use the INTC in software mode, the compiler option "-DUSE_SW_VECTOR_MODE" has to be added to the list of compiler options.
3. NO external wake-up is supported by drivers. To wake-up the core from Standby 0 over LIN channels, the ICU drivers have to be used.

5.4 ISR Macro

MCAL drivers use the ISR macro to define the functions that will process hardware interrupts. Depending on whether the OS is used or not, this macro can have different definitions:

a. OS is not used - AUTOSAR_OS_NOT_USED is defined:

i. If USE_SW_VECTOR_MODE is defined:

```
#define ISR(IsrName) void IsrName(void)
```

In this case, drivers' interrupt handlers are normal C functions and the prolog/epilog handle the context save and restore.

ii. If USE_SW_VECTOR_MODE is not defined:

```
#define ISR(IsrName) INTERRUPT_FUNC void IsrName(void)
```

In this case, drivers' interrupt handlers must save and restore the execution context.

Custom OS is used - AUTOSAR_OS_NOT_USED is not defined

```
#define ISR(IsrName) void OS_isr_##IsrName()
```

In this case, OS is handling the execution context when an interrupt occurs. Drivers' interrupt handlers are normal C functions.

Other vendor's OS is used - AUTOSAR_OS_NOT_USED is not defined. Please refer to the OS documentation for description of the ISR macro.

5.5 Other AUTOSAR modules - dependencies

- **Det** This module is necessary for enabling Development error detection. The API function used is Det_ReportError(). The activation/deactivation of Development error detection is configurable using 'LinDevErrorDetect' configuration parameter.

- **Dem:** This module is necessary for enabling reporting of production relevant error status. The API function used is Dem_ReportErrorStatus().
- **EcuM:** This module is necessary for a reference to the Wakeup source for this controller as defined in the ECU State Manager.
- **LinIf:** This module is required because it contains the definition of the NetworkHandleType type and the implementation of the LinIf_Cbk_CheckWakeup callback which is called by the LIN driver.
- **Mcu:** MCU module shall be initialized before using LIN. This module is required for setting the "LIN Clock Reference" value.
- **Port:** For each LINFlex, the TXD and RXD signals need to be configured.
- **Resource:** Sub-Derivative model is selected from Resource configuration.

Chapter 6

Main API Requirements

6.1 Main functions calls within BSW scheduler

None.

6.2 API Requirements

Not Applicable.

6.3 Calls to Notification Functions, Callbacks, Callouts

Call-back Notifications:

The LIN Driver uses 2 call back function which have to be provided by the respective module:

EcuM_SetWakeupEvent() and EcuM_CheckWakeup() have to be provided by the EcuM module.

User Notification

None

Chapter 7

Memory Allocation

7.1 Sections to be defined in MemMap.h

For Post Build data:

```
#ifdef LIN_START_CONFIG_DATA_UNSPECIFIED
#undef LIN_START_CONFIG_DATA_UNSPECIFIED
#undef MEMMAP_ERROR
/*Memory Section for Post Build Data to be defined here.
Example given in the next line*/
#pragma ghs section const=".pblin_cfg"
#endif
#ifdef LIN_STOP_CONFIG_DATA_UNSPECIFIED
#undef LIN_STOP_CONFIG_DATA_UNSPECIFIED
#undef MEMMAP_ERROR
/*End of section to be mentioned here. Example given in the
next line.*/
#pragma ghs section
#endif
```

For Code:

```
#ifdef LIN_START_SEC_CODE
#undef LIN_START_SEC_CODE
#undef MEMMAP_ERROR
/*Memory Section for Code to be defined here.*/
#endif
#ifdef LIN_STOP_SEC_CODE
#undef LIN_STOP_SEC_CODE
#undef MEMMAP_ERROR
/*End of section to be mentioned here*/
#endif
```

For Variables:

```
#ifdef LIN_START_SEC_VAR_UNSPECIFIED
#undef LIN_START_SEC_VAR_UNSPECIFIED
#undef MEMMAP_ERROR
/*Memory Section for Variables to be defined here.*/
#endif
```

Linker command file

```
#ifndef LIN_STOP_SEC_VAR_UNSPECIFIED
#undef LIN_STOP_SEC_VAR_UNSPECIFIED
#undef MEMMAP_ERROR
/*End of section to be mentioned here*/
#endif
```

For Constant data:

```
        #ifndef LIN_START_SEC_CONST_UNSPECIFIED
#undef LIN_START_SEC_CONST_UNSPECIFIED
#undef MEMMAP_ERROR
/*Memory Section for Constants to be defined here.*/
#endif
#ifndef LIN_STOP_SEC_CONST_UNSPECIFIED
#undef LIN_STOP_SEC_CONST_UNSPECIFIED
#undef MEMMAP_ERROR
/*End of section to be mentioned here*/
#endif
```

7.2 Linker command file

Memory shall be allocated for every section defined in MemMap.h.

Chapter 8

Configuration parameters considerations

Configuration parameter class for Autosar Lin driver fall into the following variants as defined below:

8.1 Configuration Parameters

Specifies whether the configuration parameter shall be of configuration class Post Build

Table 8-1. Configuration Parameters

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
Lin	IMPLEMENTATION_CONFIG_VARIANT	Pre Compile parameter for all Variants of Configuration	Pre compile
NonAutosar			
	LinEnableDualClockMode	Pre Compile parameter for all Variants of Configuration	Pre compile
	LinDisableDemReportErrorStatus	Pre Compile parameter for all Variants of Configuration	Pre compile
	LinEnablePhysicalErrorNotification	Pre Compile parameter for all Variants of Configuration	Pre compile
	LinPhysicalErrorNotificationFunction	Pre Compile parameter for all Variants of Configuration	Pre compile
LinGeneral			
	LinDevErrorDetect	Pre Compile parameter for all Variants of Configuration	Pre compile
	LinIndex	Pre Compile parameter for all Variants of Configuration	Pre compile
	LinTimeoutDuration	Pre Compile parameter for all Variants of Configuration	Pre compile
	LinVersionInfoApi	Pre Compile parameter for all Variants of Configuration	Pre compile
LinChannel			
	LinChannelId	VariantPC or VariantPB	VariantPC or VariantPB
	LinHwChannel	VariantPC or VariantPB	VariantPC or VariantPB

Table continues on the next page...

Table 8-1. Configuration Parameters (continued)

Configuration Container	Configuration Parameters	Configuration Variant	Current Implementation
	LinClockRef	VariantPC or VariantPB	VariantPC or VariantPB
	LinClockRef_Alternate	VariantPC or VariantPB	VariantPC or VariantPB
	LinChannelBaudRate	VariantPC or VariantPB	VariantPC or VariantPB
	BreakLength	VariantPC or VariantPB	VariantPC or VariantPB
	LinChannelWakeupSupport	VariantPC or VariantPB	VariantPC or VariantPB
	LinChannelEcuMWakeupSource	VariantPC or VariantPB	VariantPC or VariantPB
LinDemEventParameterRefs			
	LIN_E_TIMEOUT	VariantPC or VariantPB	VariantPC or VariantPB

Chapter 9

Integration Steps

This section gives a brief overview of the steps needed for integrating Lin Driver :

- Generate the required Lin configurations. For more details refer to section [Files required for Compilation](#)
- Allocate proper memory sections in MemMap.h and linker command file. For more details refer to section [Sections to be defined in MemMap.h](#)
- Compile & build the Lin with all the dependent modules. For more details refer to section [Building the Driver](#)



Chapter 10

ISR Reference

ISR functions exported by the Lin driver.

10.1 Software specification

The following sections contains driver software specifications.

10.1.1 Define Reference

Constants supported by the Lin driver.

10.1.2 Macros Reference

Macros supported by the Lin driver.

10.1.2.1 Macro LIN_GET_BASE_ADDRESS

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access base addresses of LINFlex IPs.

Violates: include statements in a file should only be preceded by other preprocessor directives or comments.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Prototype: `LIN_GET_BASE_ADDRESS(offset)`

10.1.2.2 Macro LINFLEX_BDRL

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register Buffer Data Register Least Significant (BDRL).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_BDRL(offset)`

10.1.2.3 Macro LINFLEX_BDRM

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register Buffer Data Register Most Significant (BDRM).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_BDRM(offset)`

10.1.2.4 Macro LINFLEX_BIDR

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register Buffer Identifier Register (BIDR).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_BIDR(offset)`

10.1.2.5 Macro LINFLEX_EXIT_NORMAL_MODE_TO_INIT_MODE

LINCR1 registers - Exit Normal Mode Enters Init Mode.

Details:

Exit the Normal mode and enters the Sleep mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Violates: C macros shall only expand to a braced initializer...

Prototype: LINFLEX_EXIT_NORMAL_MODE_TO_INIT_MODE(offset)

10.1.2.6 Macro LINFLEX_LINCFR

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register LIN Checksum Field Register (LINCFR).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LINCFR(offset)

10.1.2.7 Macro LINFLEX_LINCR1

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register LIN Control Register 1 (LINCR1).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LINCR1(offset)`

10.1.2.8 Macro LINFLEX_LINCR2

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register LIN Control Register 2 (LINCR2).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LINCR2(offset)`

10.1.2.9 Macro LINFLEX_LINESR

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register LIN Error Status Register (LINESR).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LINESR(offset)`

10.1.2.10 Macro LINFLEX_LINFBRR

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register LIN Fractional Baud Rate Register (LINFBRR).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LINFBRR(offset)`

10.1.2.11 Macro LINFLEX_LINIBRR

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register LIN Integer Baud Rate Register (LINIBRR).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LINIBRR(offset)

10.1.2.12 Macro LINFLEX_LINIER

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register LIN Interrupt Enable Register (LINIER).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LINIER(offset)

10.1.2.13 Macro LINFLEX_LINOCR

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register LIN Output Compare Register (LINOCR).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LINOCR(offset)

10.1.2.14 Macro LINFLEX_LINSR

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register LIN Status Register (LINSR).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LINSR(offset)

10.1.2.15 Macro LINFLEX_LINTCSR

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register LIN Time-Out Control Status Register (LINTCSR).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LINTCSR(offset)

10.1.2.16 Macro LINFLEX_LINTOCR

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register LIN Time-Out Control Register (LINTOCR).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LINTOCR(offset)

10.1.2.17 Macro LINFLEX_LLM_7_BITS_DATA_PARITY_Set

UARTCR registers - Word Length in UART mode.

Details:

Set the 8-bit data (or 9-bit if PCE is set) in UART mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_7_BITS_DATA_PARITY_Set(offset)`

10.1.2.18 Macro LINFLEX_LLM_8_BITS_DATA_PARITY_Set

UARTCR registers - Word Length in UART mode.

Details:

Set the 7-bit data + parity bit in UART mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_8_BITS_DATA_PARITY_Set(offset)`

10.1.2.19 Macro LINFLEX_LLM_ABRQ_Read

LINCR2 registers - Read Abort Request bits.

Details:

Read the Abort Request bits.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_ABRQ_Read(offset)`

10.1.2.20 Macro LINFLEX_LLM_ABRQ_Set

LINCR2 registers - Abort Request.

Details:

Abort the current transmission.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_ABRQ_Set(offset)`

10.1.2.21 Macro LINFLEX_LLM_AWUM_Disable

LINCR1 registers - Automatic Wake-Up Mode Disable.

Details:

The Sleep mode is exited on software request by clearing the SLEEP bit of the LINCR register.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_AWUM_Disable(offset)

10.1.2.22 Macro LINFLEX_LLM_AWUM_Enable

LINCR1 registers - Automatic Wake-Up Mode Enable.

Details:

The Sleep mode is exited automatically by hardware on RX dominant state detection. The SLEEP bit of the LINCR register is cleared by hardware whenever WUF bit in LINSR is set.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_AWUM_Enable(offset)

10.1.2.23 Macro LINFLEX_LLM_BDEF_Clear

LINESR registers - Break Delimiter Error Flag.

Details:

This macro clears the Break Delimiter Error Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_BDEF_Clear(offset)

10.1.2.24 Macro LINFLEX_LLM_BDEF_Read

LINESR registers - Break Delimiter Error Flag.

Details:

This macro read the bit indicating to the software that received Break Delimiter is too short (less than one bit time).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_BDEF_Read(offset)`

10.1.2.25 Macro LINFLEX_LLM_BDRL_Read

BDRL registers - Read all the BDRL register.

Details:

Read all the BDRL register (Data Bytes number 3,2,1 and 0 of the data field).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_BDRL_Read(offset)`

10.1.2.26 Macro LINFLEX_LLM_BDRL_Write

BDRL registers - Write all the BDRL register.

Details:

Write all the BDRL register (Data Bytes number 3,2,1 and 0 of the data field).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Prototype: `LINFLEX_LLM_BDRL_Write(offset,value)`

10.1.2.27 Macro LINFLEX_LLM_BDRM_Read

BDRM registers - Read all the BDRM register.

Details:

Read all the BDRM register (Data Bytes number 7,6,5 and 4 of the data field).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_BDRM_Read(offset)

10.1.2.28 Macro LINFLEX_LLM_BDRM_Write

BDRM registers - Write all the BDRM register.

Details:

Write all the BDRM register (Data Bytes number 7,6,5 and 4 of the data field).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Prototype: LINFLEX_LLM_BDRM_Write(offset,value)

10.1.2.29 Macro LINFLEX_LLM_BEFClear

LINEAR registers - Bit Error Flag.

Details:

This macro clears the Bit Error Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_BEFClear(offset)

10.1.2.30 Macro LINFLEX_LLM_BEF_Read

LINESR registers - Bit Error Flag.

Details:

This macro read the bit indicating to the software that LINFlex has detected a bit error.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_BEF_Read(offset)`

10.1.2.31 Macro LINFLEX_LLM_BF_Disable

LINCR1 registers - Bypass filter.

Details:

No interrupt if ID does not match any filter.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_BF_Disable(offset)`

10.1.2.32 Macro LINFLEX_LLM_BF_Enable

LINCR1 registers - Bypass filter.

Details:

An RX interrupt is generated on ID not matching any filter.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_BF_Enable(offset)`

10.1.2.33 Macro LINFLEX_LLM_BOF_Clear

LINESR registers - Buffer Overrun Flag.

Details:

This macro clears the Buffer Overrun Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_BOF_Clear(offset)`

10.1.2.34 Macro LINFLEX_LLM_BOF_Read

LINESR registers - Buffer Overrun Flag.

Details:

This macro read the bit indicating to the software that a new data byte is received and the buffer full flag is not cleared.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_BOF_Read(offset)`

10.1.2.35 Macro LINFLEX_LLM_CCD_Disable

LINCR1 registers - Checksum Calculation Disable.

Details:

Checksum calculation is done by hardware.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_CCD_Disable(offset)`

10.1.2.36 Macro LINFLEX_LLM_CCD_Enable

LINCR1 registers - Checksum Calculation Enable.

Details:

Checksum calculation is done by software.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_CCD_Enable(offset)`

10.1.2.37 Macro LINFLEX_LLM_CEF_Clear

LINESR registers - Checksum Error Flag.

Details:

This macro clears the Checksum Error Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_CEF_Clear(offset)`

10.1.2.38 Macro LINFLEX_LLM_CEF_Read

LINESR registers - Checksum Error Flag.

Details:

This macro read the bit indicating to the software that the received checksum does not match the hardware calculated checksum.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_CEF_Read(offset)`

10.1.2.39 Macro LINFLEX_LLM_CF_Read

LINCFR registers - Read Checksum bits.

Details:

Read the Checksum bits.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_CF_Read(offset)`

10.1.2.40 Macro LINFLEX_LLM_CF_Set

LINCFR registers - Set Header timeout value.

Details:

Set the Checksum bits.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Prototype: LINFLEX_LLM_CF_Set(offset,value)

10.1.2.41 Macro LINFLEX_LLM_CFD_Disable

LINCR1 registers - Checksum field Disable.

Details:

Checksum field is sent after the required number of data bytes is sent.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_CFD_Disable(offset)

10.1.2.42 Macro LINFLEX_LLM_CFD_Enable

LINCR1 registers - Checksum field Enable.

Details:

No checksum field is sent.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_CFD_Enable(offset)

10.1.2.43 Macro LINFLEX_LLM_CLASSIC_CKS_Set

BIDR registers - Set Classic Checksum for the frame.

Details:

Set Classic Checksum covering Data fields only. This is compatible with LIN specification 1.3 and below.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_CLASSIC_CKS_Set(offset)

10.1.2.44 Macro LINFLEX_LLM_COUNTER_VALUE_Read

LINTCSR registers - Timeout counter enable.

Details:

Timeout counter enable. OCF bit is set if an output compare event occurs.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_COUNTER_VALUE_Read(offset)

10.1.2.45 Macro LINFLEX_LLM_COUNTER_VALUE_Reset

LINTCSR registers - Timeout counter enable.

Details:

Timeout counter enable. OCF bit is set if an output compare event occurs.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_COUNTER_VALUE_Reset(offset)

10.1.2.46 Macro LINFLEX_LLM_DATA0_Clear

BDRL registers - Clear Data Byte 0.

Details:

Clear the Data byte 0 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA0_Clear(offset)`

10.1.2.47 Macro LINFLEX_LLM_DATA0_Read

BDRL registers - Read Data Byte 0.

Details:

Read the Data byte 0 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA0_Read(offset)`

10.1.2.48 Macro LINFLEX_LLM_DATA0_Set

BIDR registers - Set Data Byte 0.

Details:

Set the Data byte 0 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Violates: C macros shall only expand to a braced initializer...

Prototype: `LINFLEX_LLM_DATA0_Set(offset,value)`

10.1.2.49 Macro LINFLEX_LLM_DATA1_Clear

BDRL registers - Clear Data Byte 1.

Details:

Clear the Data byte 1 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA1_Clear(offset)`

10.1.2.50 Macro LINFLEX_LLM_DATA1_Read

BDRL registers - Read Data Byte 1.

Details:

Read the Data byte 1 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA1_Read(offset)`

10.1.2.51 Macro LINFLEX_LLM_DATA1_Set

BIDR registers - Set Data Byte 1.

Details:

Set the Data byte 1 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Violates: C macros shall only expand to a braced initializer...

Prototype: `LINFLEX_LLM_DATA1_Set(offset,value)`

10.1.2.52 Macro LINFLEX_LLM_DATA2_Clear

BDRL registers - Clear Data Byte 2.

Details:

Clear the Data byte 2 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA2_Clear(offset)`

10.1.2.53 Macro LINFLEX_LLM_DATA2_Read

BDRL registers - Read Data Byte 2.

Details:

Read the Data byte 2 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA2_Read(offset)`

10.1.2.54 Macro LINFLEX_LLM_DATA2_Set

BIDR registers - Set Data Byte 2.

Details:

Set the Data byte 2 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Violates: C macros shall only expand to a braced initializer...

Prototype: `LINFLEX_LLM_DATA2_Set(offset,value)`

10.1.2.55 Macro LINFLEX_LLM_DATA3_Clear

BDRL registers - Clear Data Byte 3.

Details:

Clear the Data byte 3 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA3_Clear(offset)`

10.1.2.56 Macro LINFLEX_LLM_DATA3_Read

BDRL registers - Read Data Byte 3.

Details:

Read the Data byte 3 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA3_Read(offset)`

10.1.2.57 Macro LINFLEX_LLM_DATA3_Set

BIDR registers - Set Data Byte 3.

Details:

Set the Data byte 3 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Violates: C macros shall only expand to a braced initializer...

Prototype: `LINFLEX_LLM_DATA3_Set(offset,value)`

10.1.2.58 Macro LINFLEX_LLM_DATA4_Clear

BDRL registers - Clear Data Byte 4.

Details:

Clear the Data byte 4 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA4_Clear(offset)`

10.1.2.59 Macro LINFLEX_LLM_DATA4_Read

BDRL registers - Read Data Byte 4.

Details:

Read the Data byte 4 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA4_Read(offset)`

10.1.2.60 Macro LINFLEX_LLM_DATA4_Set

BIDR registers - Set Data Byte 4.

Details:

Set the Data byte 4 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Violates: C macros shall only expand to a braced initializer...

Prototype: `LINFLEX_LLM_DATA4_Set(offset,value)`

10.1.2.61 Macro LINFLEX_LLM_DATA5_Clear

BDRL registers - Clear Data Byte 5.

Details:

Clear the Data byte 5 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA5_Clear(offset)`

10.1.2.62 Macro LINFLEX_LLM_DATA5_Read

BDRL registers - Read Data Byte 5.

Details:

Read the Data byte 5 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA5_Read(offset)`

10.1.2.63 Macro LINFLEX_LLM_DATA5_Set

BIDR registers - Set Data Byte 5.

Details:

Set the Data byte 5 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Violates: C macros shall only expand to a braced initializer...

Prototype: `LINFLEX_LLM_DATA5_Set(offset,value)`

10.1.2.64 Macro LINFLEX_LLM_DATA6_Clear

BDRL registers - Clear Data Byte 6.

Details:

Clear the Data byte 6 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA6_Clear(offset)`

10.1.2.65 Macro LINFLEX_LLM_DATA6_Read

BDRL registers - Read Data Byte 6.

Details:

Read the Data byte 6 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA6_Read(offset)`

10.1.2.66 Macro LINFLEX_LLM_DATA6_Set

BIDR registers - Set Data Byte 6.

Details:

Set the Data byte 6 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Violates: C macros shall only expand to a braced initializer...

Prototype: `LINFLEX_LLM_DATA6_Set(offset,value)`

10.1.2.67 Macro LINFLEX_LLM_DATA7_Clear

BDRL registers - Clear Data Byte 7.

Details:

Clear the Data byte 7 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA7_Clear(offset)`

10.1.2.68 Macro LINFLEX_LLM_DATA7_Read

BDRL registers - Read Data Byte 7.

Details:

Read the Data byte 7 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DATA7_Read(offset)`

10.1.2.69 Macro LINFLEX_LLM_DATA7_Set

BIDR registers - Set Data Byte 7.

Details:

Set the Data byte 7 of the data field.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Violates: C macros shall only expand to a braced initializer...

Prototype: `LINFLEX_LLM_DATA7_Set(offset,value)`

10.1.2.70 Macro LINFLEX_LLM_DATA_BUFF_EMPTY_Clear

LINSR registers - Data Buffer Empty Flag.

Details:

This macro clears the Data Buffer Empty Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_DATA_BUFF_EMPTY_Clear(offset)`

10.1.2.71 Macro LINFLEX_LLM_DATA_BUFF_FULL_Clear

LINSR registers - Data Buffer Full Flag.

Details:

This macro clears the Data Buffer Full Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_DATA_BUFF_FULL_Clear(offset)`

10.1.2.72 Macro LINFLEX_LLM_DATA_RX_COMPLETED_Clear

LINSR registers - Data Reception Completed Flag.

Details:

This macro clears the Data Reception Completed Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_DATA_RX_COMPLETED_Clear(offset)`

10.1.2.73 Macro LINFLEX_LLM_DATA_TX_COMPLETED_Clear

LINSR registers - Data Transmission Completed Flag.

Details:

This macro clears the Data Transmission Completed Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_DATA_TX_COMPLETED_Clear(offset)`

10.1.2.74 Macro LINFLEX_LLM_DDRQ_Set

LINCR2 registers - Data Discard Request.

Details:

Setting this bit stop data reception if the frame does not concern the node.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DDRQ_Set(offset)`

10.1.2.75 Macro LINFLEX_LLM_DFL_Clear

BIDR registers - Clear Data Field Length.

Details:

Clear the Data Field Length bits.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DFL_Clear(offset)`

10.1.2.76 Macro LINFLEX_LLM_DFL_Extended_Read

BIDR registers - Read Data Field Length for Extended frames.

Details:

Read DFL[0:5] where DFL[0:5] = Number of data bytes in the response part of the frame - 1.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DFL_Extended_Read(offset)`

10.1.2.77 Macro LINFLEX_LLM_DFL_Normal_Read

BIDR registers - Read Data Field Length for Normal frames.

Details:

Read DFL[0:2] where DFL[0:2] = Number of data bytes in the response part of the frame - 1.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DFL_Normal_Read(offset)`

10.1.2.78 Macro LINFLEX_LLM_DFL_Set

BIDR registers - Set Data Field Length.

Details:

Set the Data Field Length.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Prototype: `LINFLEX_LLM_DFL_Set(offset,value)`

10.1.2.79 Macro LINFLEX_LLM_DTRQ_Set

LINCR2 registers - Data Transmission Request.

Details:

Set this bit in Slave mode to request the transmission of the LIN Data.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_DTRQ_Set(offset)`

10.1.2.80 Macro **LINFLEX_LLM_ENHANCED_CKS_Set**

BIDR registers - Set Enhanced Checksum for the frame.

Details:

Set Enhanced Checksum covering Identifier and Data fields. This is compatible with LIN specification 2.0 and higher.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_ENHANCED_CKS_Set(offset)`

10.1.2.81 Macro **LINFLEX_LLM_ENTER_INIT_MODE**

LINCR1 registers - Enters Init Mode.

Details:

Enters the Init mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_ENTER_INIT_MODE(offset)`

10.1.2.82 Macro **LINFLEX_LLM_ENTER_SLEEP_MODE**

LINCR1 registers - Enters Sleep Mode.

Details:

Enters the Sleep mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_ENTER_SLEEP_MODE(offset)`

10.1.2.83 Macro LINFLEX_LLM_EXIT_INIT_MODE

LINCR1 registers - Exit Init Mode.

Details:

Exit the Init mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_EXIT_INIT_MODE(offset)`

10.1.2.84 Macro LINFLEX_LLM_EXIT_SLEEP_MODE

LINCR1 registers - Exit Sleep Mode.

Details:

Exit the Sleep mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_EXIT_SLEEP_MODE(offset)`

10.1.2.85 Macro LINFLEX_LLM_FEF_Clear

LINESR registers - Framing Error Flag.

Details:

This macro clears the Framing Error Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_FEF_Clear(offset)`

10.1.2.86 Macro LINFLEX_LLM_FEF_Read

LINESR registers - Framing Error Flag.

Details:

This macro read the bit indicating to the software that LINFlex has detected a framing error (invalid stop bit).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_FEF_Read(offset)

10.1.2.87 Macro LINFLEX_LLM_HEADER_RECEIVED

LINSR registers - Header Reception Flag.

Details:

This macro read the bit indicating to the software that valid header reception is completed.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_HEADER_RECEIVED(offset)

10.1.2.88 Macro LINFLEX_LLM_HEADER_RECEIVED_Clear

LINSR registers - Header Reception Flag.

Details:

This macro clears the Header Reception Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_HEADER_RECEIVED_Clear(offset)

10.1.2.89 Macro LINFLEX_LLM_HTO_Read

LINTOCR registers - Read Header timeout value.

Details:

Read the Header timeout value.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_HTO_Read(offset)

10.1.2.90 Macro LINFLEX_LLM_HTO_Set

LINTOCR registers - Set Header timeout value.

Details:

Set the Header timeout value.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Prototype: LINFLEX_LLM_HTO_Set(offset,value)

10.1.2.91 Macro LINFLEX_LLM_HTRQ_Set

LINCR2 registers - Set Header Transmission Request.

Details:

Request the transmission of the LIN header.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_HTRQ_Set(offset)

10.1.2.92 Macro LINFLEX_LLM_ID_Clear

BIDR registers - Clear Frame Identifier.

Details:

Clear the Identifier part of the identifier field without the identifier parity.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_ID_Clear(offset)`

10.1.2.93 Macro LINFLEX_LLM_ID_Read

BIDR registers - Read Frame Identifier.

Details:

Read the Identifier part of the identifier field without the identifier parity.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_ID_Read(offset)`

10.1.2.94 Macro LINFLEX_LLM_ID_Set

BIDR registers - Set Frame Identifier.

Details:

Set the Identifier part of the identifier field without the identifier parity.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Prototype: `LINFLEX_LLM_ID_Set(offset,value)`

10.1.2.95 Macro LINFLEX_LLM_IDPEF_Clear

LINESR registers - Identifier Parity Error Flag.

Details:

This macro clears the Identifier Parity Error Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_IDPEF_Clear(offset)

10.1.2.96 Macro LINFLEX_LLM_IDPEF_Read

LINESR registers - Identifier Parity Error Flag.

Details:

This macro read the bit indicating to the software that an Identifier Parity error occurred.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_IDPEF_Read(offset)

10.1.2.97 Macro LINFLEX_LLM_IS_BDRL_EMPTY

BDRL registers - Is the BDRL register empty ?

Details:

Verify if the BDRL register is empty.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Prototype: LINFLEX_LLM_IS_BDRL_EMPTY(offset)

10.1.2.98 Macro LINFLEX_LLM_IS_BDRM_EMPTY

BDRM registers - Is the BDRM register empty ?

Details:

Verify if the BDRM register is empty.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Prototype: LINFLEX_LLM_IS_BDRM_EMPTY(offset)

10.1.2.99 Macro LINFLEX_LLM_IS_DATA_BUFF_EMPTY

LINSR registers - Data Buffer Empty Flag.

Details:

This macro read the bit indicating to the software that the buffer is empty.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_IS_DATA_BUFF_EMPTY(offset)

10.1.2.100 Macro LINFLEX_LLM_IS_DATA_BUFF_FULL

LINSR registers - Data Buffer Full Flag.

Details:

This macro read the bit indicating to the software that the buffer is full.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_IS_DATA_BUFF_FULL(offset)

10.1.2.101 Macro LINFLEX_LLM_IS_DATA_RX_COMPLETED

LINSR registers - Data Reception Completed Flag.

Details:

This macro read the bit indicating to the software that the data reception is completed.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_IS_DATA_RX_COMPLETED(offset)

10.1.2.102 Macro LINFLEX_LLM_IS_DATA_TX_COMPLETED

LINSR registers - Data Transmission Completed Flag.

Details:

This macro read the bit indicating to the software that the data TX is completed.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_IS_DATA_TX_COMPLETED(offset)

10.1.2.103 Macro LINFLEX_LLM_IS_MSG_BUFFER_FREE

LINSR registers - Release Message Buffer.

Details:

This macro read the Message Buffer to verify if it's free.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_IS_MSG_BUFFER_FREE(offset)

10.1.2.104 Macro LINFLEX_LLM_IS_RX_BUSY

LINSR registers - Receiver Busy Flag.

Details:

This macro read the Receiver Busy Flag to verify if the receiver is idle or ongoing.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_IS_RX_BUSY(offset)

10.1.2.105 Macro LINFLEX_LLM_LIN_BREAK_10_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 10.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_10_BIT_Enable(offset)

10.1.2.106 Macro LINFLEX_LLM_LIN_BREAK_11_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 11.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_11_BIT_Enable(offset)

10.1.2.107 Macro LINFLEX_LLM_LIN_BREAK_12_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 12.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_12_BIT_Enable(offset)

10.1.2.108 Macro LINFLEX_LLM_LIN_BREAK_13_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 13.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_13_BIT_Enable(offset)

10.1.2.109 Macro LINFLEX_LLM_LIN_BREAK_14_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 14.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_14_BIT_Enable(offset)

10.1.2.110 Macro LINFLEX_LLM_LIN_BREAK_15_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 15.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_15_BIT_Enable(offset)

10.1.2.111 Macro LINFLEX_LLM_LIN_BREAK_16_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 16.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_16_BIT_Enable(offset)

10.1.2.112 Macro LINFLEX_LLM_LIN_BREAK_17_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 17.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_17_BIT_Enable(offset)

10.1.2.113 Macro LINFLEX_LLM_LIN_BREAK_18_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 18.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_18_BIT_Enable(offset)

10.1.2.114 Macro LINFLEX_LLM_LIN_BREAK_19_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 19.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_19_BIT_Enable(offset)

10.1.2.115 Macro LINFLEX_LLM_LIN_BREAK_20_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 20.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_20_BIT_Enable(offset)

10.1.2.116 Macro LINFLEX_LLM_LIN_BREAK_21_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 21.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_21_BIT_Enable(offset)

10.1.2.117 Macro LINFLEX_LLM_LIN_BREAK_22_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 22.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_22_BIT_Enable(offset)

10.1.2.118 Macro LINFLEX_LLM_LIN_BREAK_23_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 23.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_23_BIT_Enable(offset)

10.1.2.119 Macro LINFLEX_LLM_LIN_BREAK_36_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 36.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_36_BIT_Enable(offset)

10.1.2.120 Macro LINFLEX_LLM_LIN_BREAK_50_BIT_Enable

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to 50.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_BREAK_50_BIT_Enable(offset)

10.1.2.121 Macro LINFLEX_LLM_LIN_BREAK_LENGTH_Set

LINCR1 registers - LIN Master Break Length.

Details:

The following macro set the Break length in Master mode to the value parameter.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Prototype: `LINFLEX_LLM_LIN_BREAK_LENGTH_Set(offset,value)`

10.1.2.122 Macro LINFLEX_LLM_LIN_IRQ_BEIE_Disable

LINIER registers - Bit Error Interrupt Enable.

Details:

This macro disables the LINESR[BEF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_LIN_IRQ_BEIE_Disable(offset)`

10.1.2.123 Macro LINFLEX_LLM_LIN_IRQ_BEIE_Enable

LINIER registers - Bit Error Interrupt Enable.

Details:

This macro enables the LINESR[BEF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_LIN_IRQ_BEIE_Enable(offset)`

10.1.2.124 Macro LINFLEX_LLM_LIN_IRQ_BEIE_Read

LINIER registers - Bit Error Interrupt Enable.

Details:

This macro read the LINESR[BEF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_BEIE_Read(offset)

10.1.2.125 Macro LINFLEX_LLM_LIN_IRQ_BOIE_Disable

LINIER registers - Buffer Overrun Interrupt Enable.

Details:

This macro disables the LINESR[BOF] or UARTSR[BOF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_IRQ_BOIE_Disable(offset)

10.1.2.126 Macro LINFLEX_LLM_LIN_IRQ_BOIE_Enable

LINIER registers - Buffer Overrun Interrupt Enable.

Details:

This macro enables the LINESR[BOF] or UARTSR[BOF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_BOIE_Enable(offset)

10.1.2.127 Macro LINFLEX_LLM_LIN_IRQ_BOIE_Read

LINIER registers - Buffer Overrun Interrupt Enable.

Details:

This macro read the LINESR[BOF] or UARTSR[BOF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_BOIE_Read(offset)

10.1.2.128 Macro LINFLEX_LLM_LIN_IRQ_CEIE_Disable

LINIER registers - Checksum Error Interrupt Enable.

Details:

This macro disables the LINESR[CEF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_IRQ_CEIE_Disable(offset)

10.1.2.129 Macro LINFLEX_LLM_LIN_IRQ_CEIE_Enable

LINIER registers - Checksum Error Interrupt Enable.

Details:

This macro enables the LINESR[CEF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_CEIE_Enable(offset)

10.1.2.130 Macro LINFLEX_LLM_LIN_IRQ_CEIE_Read

LINIER registers - Checksum Error Interrupt Enable.

Details:

This macro read the LINESR[CEF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_CEIE_Read(offset)

10.1.2.131 Macro LINFLEX_LLM_LIN_IRQ_DBEIE_Disable

LINIER registers - Data Buffer Empty Interrupt Enable.

Details:

This macro disables the interrupt request generation when data buffer register is empty.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_IRQ_DBEIE_Disable(offset)

10.1.2.132 Macro LINFLEX_LLM_LIN_IRQ_DBEIE_Enable

LINIER registers - Data Buffer Empty Interrupt Enable.

Details:

This macro enables the interrupt request generation when data buffer register is empty.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_IRQ_DBEIE_Enable(offset)

10.1.2.133 Macro LINFLEX_LLM_LIN_IRQ_DBEIE_Read

LINIER registers - Data Buffer Empty Interrupt Enable.

Details:

This macro read the interrupt request generation when data buffer register is empty.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_DBEIE_Read(offset)

10.1.2.134 Macro LINFLEX_LLM_LIN_IRQ_DBFIE_Disable

LINIER registers - Data Buffer Full Interrupt Enable.

Details:

This macro disables the interrupt request generation when data buffer register is full.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_IRQ_DBFIE_Disable(offset)

10.1.2.135 Macro LINFLEX_LLM_LIN_IRQ_DBFIE_Enable

LINIER registers - Data Buffer Full Interrupt Enable.

Details:

This macro enables the interrupt request generation when data buffer register is full.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_IRQ_DBFIE_Enable(offset)

10.1.2.136 Macro LINFLEX_LLM_LIN_IRQ_DBFIE_Read

LINIER registers - Data Buffer Full Interrupt Enable.

Details:

This macro read the interrupt request generation when data buffer register is full.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_DBFIE_Read(offset)

10.1.2.137 Macro LINFLEX_LLM_LIN_IRQ_DRIE_Disable

LINIER registers - Data Reception Complete Interrupt Enable.

Details:

This macro disables the LINSR[DRF] or UARTSR[DRF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_IRQ_DRIE_Disable(offset)

10.1.2.138 Macro LINFLEX_LLM_LIN_IRQ_DRIE_Enable

LINIER registers - Data Reception Complete Interrupt Enable.

Details:

This macro enables the LINSR[DRF] or UARTSR[DRF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_DRIE_Enable(offset)

10.1.2.139 Macro LINFLEX_LLM_LIN_IRQ_DRIE_Read

LINIER registers - Data Reception Complete Interrupt Enable.

Details:

This macro read the LINSR[DRF] or UARTSR[DRF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_DRIE_Read(offset)

10.1.2.140 Macro LINFLEX_LLM_LIN_IRQ_DTIE_Disable

LINIER registers - Data Transmitted Complete Interrupt Enable.

Details:

This macro disables the LINSR[DTF] or UARTSR[DTF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_LIN_IRQ_DTIE_Disable(offset)`

10.1.2.141 Macro LINFLEX_LLM_LIN_IRQ_DTIE_Enable

LINIER registers - Data Transmitted Complete Interrupt Enable.

Details:

This macro enables the LINSR[DTF] or UARTSR[DTF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_LIN_IRQ_DTIE_Enable(offset)`

10.1.2.142 Macro LINFLEX_LLM_LIN_IRQ_DTIE_Read

LINIER registers - Data Transmitted Complete Interrupt Enable.

Details:

This macro read the LINSR[DTF] or UARTSR[DTF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_LIN_IRQ_DTIE_Read(offset)`

10.1.2.143 Macro LINFLEX_LLM_LIN_IRQ_FEIE_Disable

LINIER registers - Framing Error Interrupt Enable.

Details:

This macro disables the LINESR[FEF] or UARTSR[FEF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_IRQ_FEIE_Disable(offset)

10.1.2.144 Macro LINFLEX_LLM_LIN_IRQ_FEIE_Enable

LINIER registers - Framing Error Interrupt Enable.

Details:

This macro enables the LINESR[FEF] or UARTSR[FEF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_FEIE_Enable(offset)

10.1.2.145 Macro LINFLEX_LLM_LIN_IRQ_FEIE_Read

LINIER registers - Framing Error Interrupt Enable.

Details:

This macro read the LINESR[FEF] or UARTSR[FEF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_FEIE_Read(offset)

10.1.2.146 Macro LINFLEX_LLM_LIN_IRQ_HEIE_Disable

LINIER registers - Header Error Interrupt Enable.

Details:

This macro disables LINESR[BDEF], LINESR[SFEF], LINESR[IDPEF] interrupt requests generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_LIN_IRQ_HEIE_Disable(offset)`

10.1.2.147 Macro LINFLEX_LLM_LIN_IRQ_HEIE_Enable

LINIER registers - Header Error Interrupt Enable.

Details:

This macro enables the LINESR[BDEF], LINESR[SFEF], LINESR[IDPEF] interrupt requests generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_LIN_IRQ_HEIE_Enable(offset)`

10.1.2.148 Macro LINFLEX_LLM_LIN_IRQ_HEIE_Read

LINIER registers - Header Error Interrupt Enable.

Details:

This macro read the LINESR[BDEF], LINESR[SFEF], LINESR[IDPEF] interrupt requests generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_LIN_IRQ_HEIE_Read(offset)`

10.1.2.149 Macro LINFLEX_LLM_LIN_IRQ_HRIE_Disable

LINIER registers - Header Received Interrupt Enable.

Details:

This macro disables the LINSR[HRF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_LIN_IRQ_HRIE_Disable(offset)`

10.1.2.150 Macro LINFLEX_LLM_LIN_IRQ_HRIE_Enable

LINIER registers - Header Received Interrupt Enable.

Details:

This macro enables the LINSR[HRF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_LIN_IRQ_HRIE_Enable(offset)`

10.1.2.151 Macro LINFLEX_LLM_LIN_IRQ_HRIE_Read

LINIER registers - Header Received Interrupt Enable.

Details:

This macro read the LINSR[HRF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_LIN_IRQ_HRIE_Read(offset)`

10.1.2.152 Macro LINFLEX_LLM_LIN_IRQ_LSIE_Disable

LINIER registers - LIN State Interrupt Enable.

Details:

This macro disables the LIN state change (LINSR) interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_LIN_IRQ_LSIE_Disable(offset)`

10.1.2.153 Macro LINFLEX_LLM_LIN_IRQ_LSIE_Enable

LINIER registers - LIN State Interrupt Enable.

Details:

This macro enables the LIN state change (LINSR) interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_LSIE_Enable(offset)

10.1.2.154 Macro LINFLEX_LLM_LIN_IRQ_LSIE_Read

LINIER registers - LIN State Interrupt Enable.

Details:

This macro read the LIN state change (LINSR) interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_LSIE_Read(offset)

10.1.2.155 Macro LINFLEX_LLM_LIN_IRQ_OCIE_Disable

LINIER registers - Output Compare Interrupt Enable.

Details:

This macro disables the LINESR[OCF] or UARTSR[OCF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_IRQ_OCIE_Disable(offset)

10.1.2.156 Macro LINFLEX_LLM_LIN_IRQ_OCIE_Enable

LINIER registers - Output Compare Interrupt Enable.

Details:

This macro enables the LINESR[OCF] or UARTSR[OCF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_OCIE_Enable(offset)

10.1.2.157 Macro LINFLEX_LLM_LIN_IRQ_OCIE_Read

LINIER registers - Output Compare Interrupt Enable.

Details:

This macro read the LINESR[OCF] or UARTSR[OCF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_OCIE_Read(offset)

10.1.2.158 Macro LINFLEX_LLM_LIN_IRQ_SZIE_Disable

LINIER registers - Stuck at Zero Interrupt Disable.

Details:

This macro disables the LINESR[SZF] or UARTSR[SZF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_IRQ_SZIE_Disable(offset)

10.1.2.159 Macro LINFLEX_LLM_LIN_IRQ_SZIE_Enable

LINIER registers - Stuck at Zero Interrupt Enable.

Details:

This macro enables the LINESR[SZF] or UARTSR[SZF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_SZIE_Enable(offset)

10.1.2.160 Macro LINFLEX_LLM_LIN_IRQ_SZIE_Read

LINIER registers - Stuck at Zero Interrupt Read.

Details:

This macro read the LINESR[SZF] or UARTSR[SZF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_SZIE_Read(offset)

10.1.2.161 Macro LINFLEX_LLM_LIN_IRQ_WUIE_Disable

LINIER registers - Wake-up Interrupt Enable.

Details:

This macro disables the LINSR[WUF] or UARTSR[WUF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LIN_IRQ_WUIE_Disable(offset)

10.1.2.162 Macro LINFLEX_LLM_LIN_IRQ_WUIE_Enable

LINIER registers - Wake-up Interrupt Enable.

Details:

This macro enables the LINSR[WUF] or UARTSR[WUF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_WUIE_Enable(offset)

10.1.2.163 Macro LINFLEX_LLM_LIN_IRQ_WUIE_Read

LINIER registers - Wake-up Interrupt Enable.

Details:

This macro read the LINSR[WUF] or UARTSR[WUF] interrupt request generation.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_IRQ_WUIE_Read(offset)

10.1.2.164 Macro LINFLEX_LLM_LIN_MODE_Set

UARTCR registers - LIN mode enable.

Details:

Enables the LIN mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LIN_MODE_Set(offset)

10.1.2.165 Macro LINFLEX_LLM_LIN_STATE_Read

LINSR registers - LIN state.

Details:

This macro returns the LIN State: 0 Sleep mode: LINFlex is in Sleep mode to save power consumption 1 Initialization mode: LINFlex is in Initialization mode 2 Idle mode: This state is entered on several events: SLEEP bit and INIT in LINCR1 register have been cleared by software, A falling edge has been received on RX pin and AWUM bit is set,

The previous frame reception or transmission has been completed or aborted 3 Break transmission ongoing 4 Break transmission has been completed. Break Delimiter transmission is ongoing 5 Synch Field transmission is ongoing 6 Identifier transmission is ongoing 7 Header transmission is completed 8 Response reception/transmission is ongoing 9 Data reception/transmission completed. Checksum RX/TX ongoing F Sleep mode: Reset LIN State Interrupt (generated on LIN state change)

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_LIN_STATE_Read(offset)`

10.1.2.166 Macro LINFLEX_LLM_LIN_TIMEOUT_MODE_Set

LINTCSR registers - Output compare mode.

Details:

Set the Output compare mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_LIN_TIMEOUT_MODE_Set(offset)`

10.1.2.167 Macro LINFLEX_LLM_LINCR1_Reset

LINFLEX registers - LINCR1 register bit masks.

Details:

Reset LINCR1 macro.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_LINCR1_Reset(offset)`

10.1.2.168 Macro LINFLEX_LLM_LINTCSR_Reset

LINFLEX registers - LINTCSR register bit masks.

Details:

Reset LINTCSR macro.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_LINTCSR_Reset(offset)

10.1.2.169 Macro LINFLEX_LLM_LOOP_BACK_MODE_Enable

LINCR1 registers - Loop Back mode enable.

Details:

Enable the Loop Back mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_LOOP_BACK_MODE_Enable(offset)

10.1.2.170 Macro LINFLEX_LLM_MASTER_Enable

LINCR1 registers - Master Mode Enable.

Details:

Enable the Master mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_MASTER_Enable(offset)

10.1.2.171 Macro LINFLEX_LLM_NF_Clear

LINESR registers - Noise Flag.

Details:

This macro clears the Noise Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_NF_Clear(offset)`

10.1.2.172 Macro LINFLEX_LLM_NF_Read

LINESR registers - Noise Flag.

Details:

This macro read the bit indicating to the software that a noise is detected onr a received character.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_NF_Read(offset)`

10.1.2.173 Macro LINFLEX_LLM_NOT_RESET_STATE_MACHINE_ON_IOBE_Set

LINCR2 registers - LIN state machine does not reset on Bit Error.

Details:

Bit error does not reset LIN state machine.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: The compiler/linker shall be checked to ensure that 31 character signifiace and case sensitivity are supported for external identifiers.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_NOT_RESET_STATE_MACHINE_ON_IOBE_Set(offset)`

10.1.2.174 Macro LINFLEX_LLM_NOT_RESET_STATE_MACHINE_ON_IOPE_Set

LINCR2 registers - LIN state machine does not reset on Identifier Parity Error.

Details:

Identifier Parity error does not reset LIN state machine.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: The compiler/linker shall be checked to ensure that 31 character significance and case sensitivity are supported for external identifiers.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_NOT_RESET_STATE_MACHINE_ON_IOPE_Set(offset)`

10.1.2.175 Macro **LINFLEX_LLM_NOT_RESET_STATE_MACHINE_TO_IOT_Set**

LINTCSR registers - Idle on Timeout.

Details:

LIN state machine not reset to Idle on timeout event.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_NOT_RESET_STATE_MACHINE_TO_IOT_Set(offset)`

10.1.2.176 Macro **LINFLEX_LLM_OC1_Read**

LINOCR registers - Read Output compare 1 value.

Details:

Read the Output compare 1 value.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_OC1_Read(offset)`

10.1.2.177 Macro **LINFLEX_LLM_OC1_Set**

LINOCR registers - Set Output compare 1 value.

Details:

Set the Output compare 1 value.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Prototype: `LINFLEX_LLM_OC1_Set(offset,value)`

10.1.2.178 Macro LINFLEX_LLM_OC2_Read

LINOCR registers - Read Output compare 2 value.

Details:

Read the Output compare 2 value.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_OC2_Read(offset)`

10.1.2.179 Macro LINFLEX_LLM_OC2_Set

LINOCR registers - Set Output compare 2 value.

Details:

Set the Output compare 2 value.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Prototype: `LINFLEX_LLM_OC2_Set(offset,value)`

10.1.2.180 Macro LINFLEX_LLM_OCF_Clear

LINESR registers - Output Compare Flag.

Details:

This macro clears the Output Compare Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_OCF_Clear(offset)`

10.1.2.181 Macro LINFLEX_LLM_OCF_Read

LINESR registers - Output Compare Flag.

Details:

This macro read the OCF bit. If this bit is set and IOT bit in LINTCSR is set, LINFlex moves to Idle state.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_OCF_Read(offset)`

10.1.2.182 Macro LINFLEX_LLM_OUTPUT_COMPARE_MODE_Set

LINTCSR registers - LIN timeout mode.

Details:

Set the LIN timeout mode (header, response and frame timeout detection).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_OUTPUT_COMPARE_MODE_Set(offset)`

10.1.2.183 Macro LINFLEX_LLM_RELEASE_MESSAGE_BUFFER

LINSR registers - Release Message Buffer.

Details:

This macro releases the Message Buffer.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_RELEASE_MESSAGE_BUFFER(offset)`

10.1.2.184 Macro LINFLEX_LLM_RESET_BAUDRATE

LINFBRR/LINIBRR registers - Set baudrate macro.

Details:

It has a limitation for LINIBRR value on 16 bits only On Leopard this value is on 20 bits, but considering LIN 2.x standard and maximum clock rate for peripherals (120 MHz) this value cannot exceed 16 bits In case this limitation has to be avoided, also plugins modules have to be modified.

Prototype: `LINFLEX_LLM_RESET_BAUDRATE(offset)`

10.1.2.185 Macro LINFLEX_LLM_RESET_STATE_MACHINE_ON_IOBE_Set

LINCR2 registers - LIN state machine reset on Bit Error.

Details:

Bit error reset LIN state machine.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: The compiler/linker shall be checked to ensure that 31 character significance and case sensitivity are supported for external identifiers.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_RESET_STATE_MACHINE_ON_IOBE_Set(offset)`

10.1.2.186 Macro **LINFLEX_LLM_RESET_STATE_MACHINE_ON_IOPE_Set**

LINCR2 registers - LIN state machine reset on Identifier Parity Error.

Details:

Identifier Parity error reset LIN state machine.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: The compiler/linker shall be checked to ensure that 31 character significance and case sensitivity are supported for external identifiers.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_RESET_STATE_MACHINE_ON_IOPE_Set(offset)`

10.1.2.187 Macro **LINFLEX_LLM_RESET_STATE_MACHINE_TO_IOT_Set**

LINTCSR registers - Idle on Timeout.

Details:

LIN state machine reset to Idle on timeout event.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_RESET_STATE_MACHINE_TO_IOT_Set(offset)`

10.1.2.188 Macro **LINFLEX_LLM_RTO_Read**

LINTOCR registers - Read Response timeout value.

Details:

Read the Response timeout value.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_RTO_Read(offset)`

10.1.2.189 Macro LINFLEX_LLM_RTO_Set

LINTOCR registers - Set Response timeout value.

Details:

Set the Response timeout value.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Prototype: `LINFLEX_LLM_RTO_Set(offset,value)`

10.1.2.190 Macro LINFLEX_LLM_RX_BUFF_1_BYTE_Set

UARTCR registers - Transmitter Data Field length.

Details:

The following macro set the number of bytes to be received in UART mode to 1.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_RX_BUFF_1_BYTE_Set(offset)`

10.1.2.191 Macro LINFLEX_LLM_RX_BUFF_2_BYTES_Set

UARTCR registers - Transmitter Data Field length.

Details:

The following macro set the number of bytes to be received in UART mode to 2.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_RX_BUFF_2_BYTES_Set(offset)`

10.1.2.192 Macro LINFLEX_LLM_RX_BUFF_3_BYTES_Set

UARTCR registers - Transmitter Data Field length.

Details:

The following macro set the number of bytes to be received in UART mode to 3.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_RX_BUFF_3_BYTES_Set(offset)

10.1.2.193 Macro LINFLEX_LLM_RX_BUFF_4_BYTES_Set

UARTCR registers - Transmitter Data Field length.

Details:

The following macro set the number of bytes to be received in UART mode to 4.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_RX_BUFF_4_BYTES_Set(offset)

10.1.2.194 Macro LINFLEX_LLM_RX_BUSY_Clear

LINSR registers - Receiver Busy Flag.

Details:

This macro clears the Receiver Busy Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_RX_BUSY_Clear(offset)

10.1.2.195 Macro LINFLEX_LLM_RX_DIRECTION_Set

BIDR registers - Set Rx direction of the frame.

Details:

LINFLEX receives the data and copy them in the BDRL and BDRM registers.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_RX_DIRECTION_Set(offset)`

10.1.2.196 Macro LINFLEX_LLM_RX_PIN_STATE_Read

LINSR registers - LIN receive pin state.

Details:

This macro read the bit reflecting the current status of LINRX pin for diagnostic purposes.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_RX_PIN_STATE_Read(offset)`

10.1.2.197 Macro LINFLEX_LLM_SELF_TEST_MODE_Enable

LINCR1 registers - Self Test mode enable.

Details:

Enable the Self Test mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Violates: Violates MISRA 2004 Required Rule 19.10, arguments shall be enclosed in parentheses.

Violates: C macros shall only expand to a braced initializer...

Prototype: `LINFLEX_LLM_SELF_TEST_MODE_Enable(offset)`

10.1.2.198 Macro LINFLEX_LLM_SET_BAUDRATE

Prototype: LINFLEX_LLM_SET_BAUDRATE(offset,value)

10.1.2.199 Macro LINFLEX_LLM_SFEF_Clear

LINESR registers - Synch Field Error Flag.

Details:

This macro clears the Synch Field Error Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_SFEF_Clear(offset)

10.1.2.200 Macro LINFLEX_LLM_SFEF_Read

LINESR registers - Synch Field Error Flag.

Details:

This macro read the bit indicating to the software that Synch Field error occurred (inconsistent Synch Field).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_SFEF_Read(offset)

10.1.2.201 Macro LINFLEX_LLM_SLAVE_Enable

LINCR1 registers - Slave Mode Enable.

Details:

Enable the Slave mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_SLAVE_Enable(offset)

10.1.2.202 Macro LINFLEX_LLM_SZF_Clear

LINESR registers - Stuck at zero Flag.

Details:

This macro clears the Stuck at zero Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_SZF_Clear(offset)

10.1.2.203 Macro LINFLEX_LLM_SZF_Read

LINESR registers - Stuck at zero Flag.

Details:

This macro read the bit indicating to the software that the bus is dominant for more than a 100-bit time.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_SZF_Read(offset)

10.1.2.204 Macro LINFLEX_LLM_TIMEOUT_COUNTER_Disable

LINTCSR registers - Timeout counter disable.

Details:

Timeout counter disable. OCF bit in LINESR or UARTSR is not set on an output compare event

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_TIMEOUT_COUNTER_Disable(offset)

10.1.2.205 Macro LINFLEX_LLM_TIMEOUT_COUNTER_Enable

LINTCSR registers - Timeout counter enable.

Details:

Timeout counter enable. OCF bit is set if an output compare event occurs.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: LINFLEX_LLM_TIMEOUT_COUNTER_Enable(offset)

10.1.2.206 Macro LINFLEX_LLM_TX_BUFF_1_BYTE_Set

UARTCR registers - Transmitter Data Field length.

Details:

The following macro set the number of bytes to be transmitted in UART mode to 1.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_TX_BUFF_1_BYTE_Set(offset)

10.1.2.207 Macro LINFLEX_LLM_TX_BUFF_2_BYTES_Set

UARTCR registers - Transmitter Data Field length.

Details:

The following macro set the number of bytes to be transmitted in UART mode to 2.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_TX_BUFF_2_BYTES_Set(offset)

10.1.2.208 Macro LINFLEX_LLM_TX_BUFF_3_BYTES_Set

UARTCR registers - Transmitter Data Field length.

Details:

The following macro set the number of bytes to be transmitted in UART mode to 3.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_TX_BUFF_3_BYTES_Set(offset)

10.1.2.209 Macro LINFLEX_LLM_TX_BUFF_4_BYTES_Set

UARTCR registers - Transmitter Data Field length.

Details:

The following macro set the number of bytes to be transmitted in UART mode to 4.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_TX_BUFF_4_BYTES_Set(offset)

10.1.2.210 Macro LINFLEX_LLM_TX_DIRECTION_Set

BIDR registers - Set Tx direction of the frame.

Details:

LINFLEX transmits the data from the BDR registers.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_TX_DIRECTION_Set(offset)

10.1.2.211 Macro LINFLEX_LLM_UART_BOF_Clear

UARTSR registers - Buffer Overrun Flag.

Details:

This macro clears the Buffer Overrun Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_BOF_Clear(offset)`

10.1.2.212 Macro LINFLEX_LLM_UART_BOF_Read

UARTSR registers - Buffer Overrun Flag.

Details:

This macro read the bit indicating to the software that a new data byte is received and the buffer full flag is not cleared.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_BOF_Read(offset)`

10.1.2.213 Macro LINFLEX_LLM_UART_DRF_Clear

UARTSR registers - Data Reception Completed Flag.

Details:

This macro clears the Data Reception Completed Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_DRF_Clear(offset)`

10.1.2.214 Macro LINFLEX_LLM_UART_DRF_Read

UARTSR registers - Data Reception Completed Flag.

Details:

This macro read the bit indicating to the software that the data reception is completed.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_DRF_Read(offset)`

10.1.2.215 Macro LINFLEX_LLM_UART_DTF_Clear

UARTSR registers - Data Transmission Completed Flag.

Details:

This macro clears the Data Transmission Completed Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_DTF_Clear(offset)`

10.1.2.216 Macro LINFLEX_LLM_UART_DTF_Read

UARTSR registers - Data Transmission Completed Flag.

Details:

This macro read the bit indicating to the software that the data transmission is completed.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_DTF_Read(offset)`

10.1.2.217 Macro LINFLEX_LLM_UART_EVEN_PARITY_Set

UARTCR registers - Even Parity.

Details:

Set the Even Parity in UART mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_UART_EVEN_PARITY_Set(offset)`

10.1.2.218 Macro LINFLEX_LLM_UART_FEF_Clear

UARTSR registers - Framing Error Flag.

Details:

This macro clears the Framing Error Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_UART_FEF_Clear(offset)

10.1.2.219 Macro LINFLEX_LLM_UART_FEF_Read

UARTSR registers - Framing Error Flag.

Details:

This macro read the bit indicating to the software that LINFlex has detected a framing error (invalid stop bit).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_UART_FEF_Read(offset)

10.1.2.220 Macro LINFLEX_LLM_UART_MODE_Set

UARTCR registers - UART mode enable.

Details:

Enables the UART mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_UART_MODE_Set(offset)

10.1.2.221 Macro LINFLEX_LLM_UART_NF_Clear

UARTSR registers - Noise Flag.

Details:

This macro clears the Noise Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_UART_NF_Clear(offset)

10.1.2.222 Macro LINFLEX_LLM_UART_NF_Read

UARTSR registers - Noise Flag.

Details:

This macro read the bit indicating to the software that noise is detected on a received character.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_UART_NF_Read(offset)

10.1.2.223 Macro LINFLEX_LLM_UART_OCF_Clear

UARTSR registers - Output Compare Flag.

Details:

This macro clears the Output Compare Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_UART_OCF_Clear(offset)

10.1.2.224 Macro LINFLEX_LLM_UART_OCF_Read

UARTSR registers - Output Compare Flag.

Details:

This macro read the bit indicating to the software that the counter has matched the content of OC1[0:7] or OC2[0:7] in LINOCR.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_OCF_Read(offset)`

10.1.2.225 Macro LINFLEX_LLM_UART_ODD_PARITY_Set

UARTCR registers - Odd Parity.

Details:

Set the Odd Parity in UART mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_ODD_PARITY_Set(offset)`

10.1.2.226 Macro LINFLEX_LLM_UART_PARITY_CONTROL_Disable

UARTCR registers - Parity Control Disable.

Details:

Disable the Parity transmit/check in UART mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: The compiler/linker shall be checked to ensure that 31 character significance and case sensitivity are supported for external identifiers.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_UART_PARITY_CONTROL_Disable(offset)`

10.1.2.227 Macro LINFLEX_LLM_UART_PARITY_CONTROL_Enable

UARTCR registers - Parity Control Enable.

Details:

Enable the Parity transmit/check in UART mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Violates: This is required as per Autosar Requirement.

Prototype: `LINFLEX_LLM_UART_PARITY_CONTROL_Enable(offset)`

10.1.2.228 Macro LINFLEX_LLM_UART_PE0_Clear

UARTSR registers - Parity Error Flag Rx0.

Details:

This macro clears the Parity Error Flag Rx0.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_PE0_Clear(offset)`

10.1.2.229 Macro LINFLEX_LLM_UART_PE0_Read

UARTSR registers - Parity Error Flag Rx0.

Details:

This macro read the bit indicating to the software that a parity error in the corresponding received byte (Rx0).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_PE0_Read(offset)`

10.1.2.230 Macro LINFLEX_LLM_UART_PE1_Clear

UARTSR registers - Parity Error Flag Rx1.

Details:

This macro clears the Parity Error Flag Rx1.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_PE1_Clear(offset)`

10.1.2.231 Macro LINFLEX_LLM_UART_PE1_Read

UARTSR registers - Parity Error Flag Rx1.

Details:

This macro read the bit indicating to the software that a parity error in the corresponding received byte (Rx1).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_PE1_Read(offset)`

10.1.2.232 Macro LINFLEX_LLM_UART_PE2_Clear

UARTSR registers - Parity Error Flag Rx2.

Details:

This macro clears the Parity Error Flag Rx2.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_PE2_Clear(offset)`

10.1.2.233 Macro LINFLEX_LLM_UART_PE2_Read

UARTSR registers - Parity Error Flag Rx2.

Details:

This macro read the bit indicating to the software that a parity error in the corresponding received byte (Rx2).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_PE2_Read(offset)`

10.1.2.234 Macro LINFLEX_LLM_UART_PE3_Clear

UARTSR registers - Parity Error Flag Rx3.

Details:

This macro clears the Parity Error Flag Rx3.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_PE3_Clear(offset)`

10.1.2.235 Macro LINFLEX_LLM_UART_PE3_Read

UARTSR registers - Parity Error Flag Rx3.

Details:

This macro read the bit indicating to the software that a parity error in the corresponding received byte (Rx3).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_PE3_Read(offset)`

10.1.2.236 Macro LINFLEX_LLM_UART_RMB_Clear

UARTSR registers - Release Message Buffer.

Details:

This macro releases the Message Buffer.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_RMB_Clear(offset)`

10.1.2.237 Macro LINFLEX_LLM_UART_RMB_Read

UARTSR registers - Release Message Buffer.

Details:

This macro read the Message Buffer to verify if it's free.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_UART_RMB_Read(offset)

10.1.2.238 Macro LINFLEX_LLM_UART_RPS_Read

UARTSR registers - UART receive pin state.

Details:

This macro read the bit reflecting the current status of UARTRX pin for diagnostic purposes.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_UART_RPS_Read(offset)

10.1.2.239 Macro LINFLEX_LLM_UART_RX_Disable

UARTCR registers - Receiver Disable.

Details:

Disables the RX in UART mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_UART_RX_Disable(offset)

10.1.2.240 Macro LINFLEX_LLM_UART_RX_Enable

UARTCR registers - Receiver Enable.

Details:

Enables the RX in UART mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_UART_RX_Enable(offset)

10.1.2.241 Macro LINFLEX_LLM_UART_SZF_Clear

UARTSR registers - Stuck at zero Flag.

Details:

This macro clears the Stuck at zero Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_UART_SZF_Clear(offset)

10.1.2.242 Macro LINFLEX_LLM_UART_SZF_Read

UARTSR registers - Stuck at zero Flag.

Details:

This macro read the bit indicating to the software that bus is dominant for more than a 100-bit time.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: LINFLEX_LLM_UART_SZF_Read(offset)

10.1.2.243 Macro LINFLEX_LLM_UART_TX_Disable

UARTCR registers - Transmitter Disable.

Details:

Disables the TX in UART mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_TX_Disable(offset)`

10.1.2.244 Macro LINFLEX_LLM_UART_TX_Enable

UARTCR registers - Transmitter Enable.

Details:

Enables the TX in UART mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_TX_Enable(offset)`

10.1.2.245 Macro LINFLEX_LLM_UART_WUF_Clear

UARTSR registers - Wake-up Flag.

Details:

This macro clears the Wake-up Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_WUF_Clear(offset)`

10.1.2.246 Macro LINFLEX_LLM_UART_WUF_Read

UARTSR registers - Wake-up Flag.

Details:

This macro read the bit indicating to the software that LINFlex has detected a falling edge on the LINRX pin in Sleep mode.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_UART_WUF_Read(offset)`

10.1.2.247 Macro LINFLEX_LLM_WAKEUP_Clear

LINSR registers - Wake-up Flag.

Details:

This macro clears the Wake-up Flag.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_WAKEUP_Clear(offset)`

10.1.2.248 Macro LINFLEX_LLM_WAKEUP_RECEIVED

LINSR registers - Wake-up Flag.

Details:

This macro read the bit indicating to the software that LINFlex has detected a falling edge on the LINRX pin.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_WAKEUP_RECEIVED(offset)`

10.1.2.249 Macro LINFLEX_LLM_WURQ_Set

LINCR2 registers - Wake-up Generation Request.

Details:

Setting this bit generates a wake-up pulse.

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_LLM_WURQ_Set(offset)`

10.1.2.250 Macro LINFLEX_UARTCR

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register UART Mode Control Register (UARTCR).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_UARTCR(offset)`

10.1.2.251 Macro LINFLEX_UARTSR

LINFLEX Registers macro definition.

Details:

LINFLEX macro to access register UART Mode Status Register (UARTSR).

Violates: MISRA 2004 Advisory Rule 19.7, Function-like macro defined.

Prototype: `LINFLEX_UARTSR(offset)`

10.1.3 Enum Reference

Enumeration of all constants supported by the Lin driver.

10.1.4 Function Reference

Functions supported by the Lin driver.

10.1.4.1 Function Lin_LLD_IsrError_LINFlex_0

Error interrupt on LinFlex_0.



Figure 10-1. Function Lin_LLD_IsrError_LINFlex_0 References.

Details:

This function implements the ISR occurring on error on LinFlex_0.

Note

This interrupt handlers is only present if Hardware Channel 0 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrError_LINFlex_0(void);`

10.1.4.2 Function Lin_LLD_IsrError_LINFlex_1

Error interrupt on LinFlex_1.



Figure 10-2. Function Lin_LLD_IsrError_LINFlex_1 References.

Details:

This function implements the ISR occurring on error on LinFlex_1.

Note

This interrupt handlers is only present if Hardware Channel 1 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrError_LINFlex_1(void);`

10.1.4.3 Function Lin_LLD_IsrError_LINFlex_2

Error interrupt on LinFlex_2.



Figure 10-3. Function Lin_LLD_IsrError_LINFlex_2 References.

Details:

This function implements the ISR occurring on error on LinFlex_2.

Note

This interrupt handlers is only present if Hardware Channel 2 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrError_LINFlex_2(void);`

10.1.4.4 Function Lin_LLD_IsrError_LINFlex_3

Error interrupt on LinFlex_3.



Figure 10-4. Function Lin_LLD_IsrError_LINFlex_3 References.

Details:

This function implements the ISR occurring on error on LinFlex_3.

Note

This interrupt handlers is only present if Hardware Channel 3 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrError_LINFlex_3(void);`

10.1.4.5 Function Lin_LLD_IsrError_LINFlex_4

Error interrupt on LinFlex_4.



Figure 10-5. Function Lin_LLD_IsrError_LINFlex_4 References.

Details:

This function implements the ISR occurring on error on LinFlex_4.

Note

This interrupt handlers is only present if Hardware Channel 4 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrError_LINFlex_4(void);`

10.1.4.6 Function Lin_LLD_IsrError_LINFlex_5

Error interrupt on LinFlex_5.



Figure 10-6. Function Lin_LLD_IsrError_LINFlex_5 References.

Details:

This function implements the ISR occurring on error on LinFlex_5.

Note

This interrupt handlers is only present if Hardware Channel 5 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrError_LINFlex_5(void);`

10.1.4.7 Function Lin_LLD_IsrError_LINFlex_6

Error interrupt on LinFlex_6.



Figure 10-7. Function Lin_LLD_IsrError_LINFlex_6 References.

Details:

This function implements the ISR occurring on error on LinFlex_6.

Note

This interrupt handlers is only present if Hardware Channel 6 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrError_LINFlex_6(void);`

10.1.4.8 Function Lin_LLD_IsrError_LINFlex_7

Error interrupt on LinFlex_7.



Figure 10-8. Function Lin_LLD_IsrError_LINFlex_7 References.

Details:

This function implements the ISR occurring on error on LinFlex_7.

Note

This interrupt handlers is only present if Hardware Channel 7 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrError_LINFlex_7(void);`

10.1.4.9 Function Lin_LLD_IsrError_LINFlex_8

Error interrupt on LinFlex_8.



Figure 10-9. Function Lin_LLD_IsrError_LINFlex_8 References.

Details:

This function implements the ISR occurring on error on LinFlex_8.

Note

This interrupt handlers is only present if Hardware Channel 8 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrError_LINFlex_8(void);`

10.1.4.10 Function Lin_LLD_IsrError_LINFlex_9

Error interrupt on LinFlex_9.



Figure 10-10. Function Lin_LLD_IsrError_LINFlex_9 References.

Details:

This function implements the ISR occurring on error on LinFlex_9.

Note

This interrupt handlers is only present if Hardware Channel 9 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrError_LINFlex_9(void);`

10.1.4.11 Function Lin_LLD_IsrRx_LINFlex_0

Rx interrupt on LinFlex_0.



Figure 10-11. Function Lin_LLD_IsrRx_LINFlex_0 References.

Details:

This function implements the ISR occurring on reception on LinFlex_0.

Note

This interrupt handlers is only present if Hardware Channel 0 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrRx_LINFlex_0(void);`

10.1.4.12 Function Lin_LLD_IsrRx_LINFlex_1

Rx interrupt on LinFlex_1.



Figure 10-12. Function Lin_LLD_IsrRx_LINFlex_1 References.

Details:

This function implements the ISR occurring on reception on LinFlex_1.

Note

This interrupt handlers is only present if Hardware Channel 1 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrRx_LINFlex_1(void);`

10.1.4.13 Function Lin_LLD_IsrRx_LINFlex_2

Rx interrupt on LinFlex_2.



Figure 10-13. Function Lin_LLD_IsrRx_LINFlex_2 References.

Details:

This function implements the ISR occurring on reception on LinFlex_2.

Note

This interrupt handlers is only present if Hardware Channel 2 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrRx_LINFlex_2(void);`

10.1.4.14 Function Lin_LLD_IsrRx_LINFlex_3

Rx interrupt on LinFlex_3.



Figure 10-14. Function Lin_LLD_IsrRx_LINFlex_3 References.

Details:

This function implements the ISR occurring on reception on LinFlex_3.

Note

This interrupt handlers is only present if Hardware Channel 3 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrRx_LINFlex_3(void);`

10.1.4.15 Function Lin_LLD_IsrRx_LINFlex_4

Rx interrupt on LinFlex_4.



Figure 10-15. Function Lin_LLD_IsrRx_LINFlex_4 References.

Details:

This function implements the ISR occurring on reception on LinFlex_4.

Note

This interrupt handlers is only present if Hardware Channel 4 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrRx_LINFlex_4(void);`

10.1.4.16 Function Lin_LLD_IsrRx_LINFlex_5

Rx interrupt on LinFlex_5.



Figure 10-16. Function Lin_LLD_IsrRx_LINFlex_5 References.

Details:

This function implements the ISR occurring on reception on LinFlex_5.

Note

This interrupt handlers is only present if Hardware Channel 5 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrRx_LINFlex_5(void);`

10.1.4.17 Function Lin_LLD_IsrRx_LINFlex_6

Rx interrupt on LinFlex_6.



Figure 10-17. Function Lin_LLD_IsrRx_LINFlex_6 References.

Details:

This function implements the ISR occurring on reception on LinFlex_6.

Note

This interrupt handlers is only present if Hardware Channel 6 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrRx_LINFlex_6(void);`

10.1.4.18 Function Lin_LLD_IsrRx_LINFlex_7

Rx interrupt on LinFlex_7.



Figure 10-18. Function Lin_LLD_IsrRx_LINFlex_7 References.

Details:

This function implements the ISR occurring on reception on LinFlex_7.

Note

This interrupt handlers is only present if Hardware Channel 7 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrRx_LINFlex_7(void);`

10.1.4.19 Function Lin_LLD_IsrRx_LINFlex_8

Rx interrupt on LinFlex_8.



Figure 10-19. Function Lin_LLD_IsrRx_LINFlex_8 References.

Details:

This function implements the ISR occurring on reception on LinFlex_8.

Note

This interrupt handlers is only present if Hardware Channel 8 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrRx_LINFlex_8(void);`

10.1.4.20 Function Lin_LLD_IsrRx_LINFlex_9

Rx interrupt on LinFlex_9.



Figure 10-20. Function Lin_LLD_IsrRx_LINFlex_9 References.

Details:

This function implements the ISR occurring on reception on LinFlex_9.

Note

This interrupt handlers is only present if Hardware Channel 9 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrRx_LINFlex_9(void);`

10.1.4.21 Function Lin_LLD_IsrTx_LINFlex_0

Tx interrupt on LinFlex_0.



Figure 10-21. Function Lin_LLD_IsrTx_LINFlex_0 References.

Details:

This function implements the ISR occurring on transmission on LinFlex_0.

Note

This interrupt handlers is only present if Hardware Channel 0 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrTx_LINFlex_0(void);`

10.1.4.22 Function Lin_LLD_IsrTx_LINFlex_1

Tx interrupt on LinFlex_1.



Figure 10-22. Function Lin_LLD_IsrTx_LINFlex_1 References.

Details:

This function implements the ISR occurring on transmission on LinFlex_1.

Note

This interrupt handlers is only present if Hardware Channel 1 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrTx_LINFlex_1(void);`

10.1.4.23 Function Lin_LLD_IsrTx_LINFlex_2

Tx interrupt on LinFlex_2.



Figure 10-23. Function Lin_LLD_IsrTx_LINFlex_2 References.

Details:

This function implements the ISR occurring on transmission on LinFlex_2.

Note

This interrupt handlers is only present if Hardware Channel 2 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrTx_LINFlex_2(void);`

10.1.4.24 Function Lin_LLD_IsrTx_LINFlex_3

Tx interrupt on LinFlex_3.



Figure 10-24. Function Lin_LLD_IsrTx_LINFlex_3 References.

Details:

This function implements the ISR occurring on transmission on LinFlex_3.

Note

This interrupt handlers is only present if Hardware Channel 3 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrTx_LINFlex_3(void);`

10.1.4.25 Function Lin_LLD_IsrTx_LINFlex_4

Tx interrupt on LinFlex_4.



Figure 10-25. Function Lin_LLD_IsrTx_LINFlex_4 References.

Details:

This function implements the ISR occurring on transmission on LinFlex_4.

Note

This interrupt handlers is only present if Hardware Channel 4 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrTx_LINFlex_4(void);`

10.1.4.26 Function Lin_LLD_IsrTx_LINFlex_5

Tx interrupt on LinFlex_5.



Figure 10-26. Function Lin_LLD_IsrTx_LINFlex_5 References.

Details:

This function implements the ISR occurring on transmission on LinFlex_5.

Note

This interrupt handlers is only present if Hardware Channel 5 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrTx_LINFlex_5(void);`

10.1.4.27 Function Lin_LLD_IsrTx_LINFlex_6

Tx interrupt on LinFlex_6.



Figure 10-27. Function Lin_LLD_IsrTx_LINFlex_6 References.

Details:

This function implements the ISR occurring on transmission on LinFlex_6.

Note

This interrupt handlers is only present if Hardware Channel 6 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrTx_LINFlex_6(void);`

10.1.4.28 Function Lin_LLD_IsrTx_LINFlex_7

Tx interrupt on LinFlex_7.



Figure 10-28. Function Lin_LLD_IsrTx_LINFlex_7 References.

Details:

This function implements the ISR occurring on transmission on LinFlex_7.

Note

This interrupt handlers is only present if Hardware Channel 7 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrTx_LINFlex_7(void);`

10.1.4.29 Function Lin_LLD_IsrTx_LINFlex_8

Tx interrupt on LinFlex_8.



Figure 10-29. Function Lin_LLD_IsrTx_LINFlex_8 References.

Details:

This function implements the ISR occurring on transmission on LinFlex_8.

Note

This interrupt handlers is only present if Hardware Channel 8 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrTx_LINFlex_8(void);`

10.1.4.30 Function Lin_LLD_IsrTx_LINFlex_9

Tx interrupt on LinFlex_9.



Figure 10-30. Function Lin_LLD_IsrTx_LINFlex_9 References.

Details:

This function implements the ISR occurring on transmission on LinFlex_9.

Note

This interrupt handlers is only present if Hardware Channel 9 is used and DMA is not used.

Implements: DLIN03100, DLIN03400

Prototype: `void Lin_LLD_IsrTx_LINFlex_9(void);`

10.1.5 Structs Reference

Data structures supported by the Lin driver.

10.1.6 Types Reference

Types structures supported by the Lin driver.

10.1.7 Variables Reference

Variables supported by the Lin driver.

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